

Meaningful Differentiation in Markets with Overlapping Categories: Evidence from Fragrances

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Abstract

Consumers often value products that depart from category norms, yet the conditions under which differentiation is rewarded remain poorly understood. Prior research typically conceptualizes atypicality as the distance from a category prototype. We argue instead that the value of differentiation depends not only on the magnitude of deviation but also on its direction within a structured category space organized around multiple local prototypes. We introduce the concept of *directional atypicality* and examine it in the fragrance market, where men's and women's segments coexist with a more fluid unisex segment. Using large-scale data on fragrances and consumer reviews, we find that atypicality is associated with higher evaluations for men's and women's fragrances but lower evaluations for unisex fragrances. These relationships are largely obscured when atypicality is measured relative to a single aggregate benchmark. Mapping fragrances into a common category space shows that successful differentiation follows systematic directional movements, while consumer reviews reveal distinct patterns of recombination, balance, and integration underlying them. Our findings extend research on categorization, creativity, and category spanning, provide a large-scale investigation of consumer evaluation in the fragrance market, and offer a general framework for differentiation, innovation, and positioning in markets with multiple overlapping categories.

Keywords: Directional Atypicality; Differentiation; Categorization; Creative Recombination; Machine Learning; Text Analysis; Consumer Reviews

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1 Introduction

Consumers often value products that deviate from established category norms, yet the conditions under which such differentiation is rewarded remain poorly understood. Consider a men’s fragrance that blends sweet edible notes with traditional woody scents or a women’s fragrance that combines floral notes with leather and spices. Some consumers may view these fragrances as distinctive and innovative, whereas others may perceive them as inconsistent with category expectations. Consistent with these contrasting responses, prior research shows that deviation can enhance evaluations by increasing distinctiveness but may also reduce appeal when products become difficult to interpret (Meyers-Levy and Tybout, 1989). Related work on differentiation, identity signaling, and optimal distinctiveness likewise suggests that consumers sometimes prefer products that diverge from prevailing norms, while in other contexts favor those that remain closer to them (Askin and Mauskopf, 2017; Berger and Heath, 2007; Brewer, 1991). Research on creativity further demonstrates that successful products often recombine familiar and novel elements in ways that preserve coherence and interpretability (Burroughs, Moreau, and Mick, 2018; Goldenberg, Mazursky, and Solomon, 1999; Moreau, Lehmann, and Markman, 2001; Toubia and Netzer, 2017; Uzzi et al., 2013). Collectively, these findings suggest that consumers value some forms of differentiation more than others, yet they provide limited insight into why comparable levels of deviation are rewarded in some settings but penalized in others.

One possible explanation for these contrasting evaluations lies in how consumers organize products into categories (Cohen and Basu, 1987). Consider the men’s fragrance introduced above that blends sweet edible notes with traditional woody scents. One consumer may evaluate it relative to a traditional men’s prototype and perceive it as an unusual departure from category expectations, whereas another may evaluate it relative to a more hybrid unisex prototype and view it as an appealing reinterpretation of the category. A complementary stream of research emphasizes the role of category structure and typicality in shaping consumer evaluations (Barsalou, 1985; Blanchard et al., 2012; Loken, 2006; Loken and Ward, 1990; Ratneshwar and Shocker, 1991; Rosch, 1978). Categories are typically organized around prototypes that serve as cognitive reference points, and products are evaluated according to their similarity to these prototypes.

Existing research, however, generally conceptualizes atypicality as a scalar construct that captures only the distance between a product and a single category benchmark. Yet many markets contain multiple, overlapping categories with distinct local prototypes. Related research recognizes that consumers may simultaneously draw on multiple categories under ambiguity or categorization uncertainty (Gregan-Paxton, Hoeffler, and Zhao, 2005; Lajos et al., 2009). Our focus differs by examining how products move within a common category space defined by multiple local prototypes and the evaluative consequences of those movements. Consequently, the value of differentiation depends not only on how far products depart from category norms, but also on where those departures position them within the broader category structure. This raises a fundamental question: when multiple local prototypes coexist, how does category structure shape the value of differentiation?

We investigate this question in the fragrance market, where gender-based segmentation creates well-defined categories (men’s and women’s fragrances) alongside a more fluid unisex category.

This combination of well-defined yet overlapping market segments and rich variation in olfactory profiles makes the fragrance market particularly well suited for studying how consumers evaluate differentiation when multiple local prototypes coexist. Using a large dataset of fragrances and consumer reviews, we find that atypicality is associated with higher evaluations for men’s and women’s fragrances but lower evaluations for unisex fragrances. Importantly, these relationships are largely obscured when atypicality is measured relative to a single aggregate benchmark. Pooling fragrances across distinct local prototypes combines deviations defined relative to different reference points, producing a weak overall relationship that masks substantial heterogeneity across segments.

To explain these findings, we introduce the concept of *directional atypicality*. We argue that the value of differentiation depends not only on how far a product departs from category norms, but also on the direction of that departure within a structured category space. Returning to the example above, the same men’s fragrance may become atypical in different ways. Adding sweet edible notes moves it toward a neighboring unisex prototype, whereas strengthening its woody, leather, and tobacco character moves it toward a more extreme masculine prototype. Although both fragrances may be equally atypical, they occupy different positions within category space. More generally, products in markets characterized by multiple local prototypes may become atypical by moving toward neighboring categories, toward more extreme versions of the focal category, or toward hybrid positions that combine elements of multiple categories. These alternative directions of differentiation can have markedly different evaluative consequences despite involving similar levels of deviation.

Consistent with this perspective, we find that as atypicality increases, men’s and women’s fragrances move toward a shared cross-segment region, whereas unisex fragrances remain anchored near a balanced central prototype. Once these directional movements are taken into account, the direct association between atypicality and consumer evaluations largely disappears. Consumer reviews provide a behavioral explanation for why these directional movements are valued differently. In the gendered segments, atypical fragrances are characterized by material-centered and cross-gender recombination, the intensification of familiar accords, and the blurring of traditional gender boundaries. In contrast, typical unisex fragrances emphasize balanced olfactory recombination, compositional integration, and coherence. Together, these findings suggest that consumers reward forms of differentiation that move products toward valued and interpretable positions within category space, rather than differentiation alone.

Our study contributes to research on categorization, schema congruity, creativity, and category spanning by demonstrating that the value of differentiation depends not only on distance from a category prototype but also on the structure of category space. We introduce the concept of *directional atypicality*, which captures how products are positioned relative to multiple local prototypes. This perspective extends prototype-based accounts of categorization by shifting attention from scalar distance to directional positioning. It also complements research on creative recombination by showing that consumers do not reward novelty or recombination in itself. Rather, successful differentiation emerges when recombination produces products that remain interpretable within existing category structures. Finally, our findings extend research on category spanning by showing that the value of crossing category boundaries depends not only on whether products span categories, but also on where those movements reposition products within the broader category system.

Although we study fragrances, the proposed framework applies more broadly to markets characterized by multiple overlapping categories, including fashion, food, media, entertainment, and other cultural goods, where products are likewise evaluated relative to multiple local prototypes. Beyond its theoretical contributions, our study also contributes to the marketing literature on fragrance products. Whereas prior marketing research has focused primarily on the effects of scent on consumer mood, perception, and retail atmospherics (Bosmans, 2006; Krishna, 2012; Spangenberg, Crowley, and Henderson, 1996), comparatively less attention has been devoted to understanding how consumers evaluate fragrance products themselves. By combining large-scale data on fragrance composition, consumer ratings, and textual reviews, our study provides new evidence on the drivers of consumer evaluations in a market where product differentiation plays a central role in consumer choice.

The remainder of the paper is organized as follows. We first describe the data and examine the relationship between atypicality and consumer evaluations. We then investigate category structure and introduce measures of directional atypicality. Next, we analyze consumer reviews to provide a behavioral interpretation of the directional patterns identified in the structured-data analysis. Finally, we discuss the theoretical and managerial implications of our findings, along with their limitations and directions for future research.

2 Data

We use a dataset collected from *Fragrantica*, an online platform where users rate and review fragrances. The dataset comprises 61,868 fragrances from 1,732 brands and 1,337 perfumers, known as *nez*. Release-year information is available for 81% of fragrances, spanning seven decades, with the 2010s accounting for the largest share (41%). Each fragrance is classified by the industry as men’s (10,786), women’s (25,468), or unisex (25,614), providing the segment definitions used throughout the analysis (Edwards, 2020). For each fragrance, we observe the average user rating and its olfactory profile, characterized by up to 90 olfactory accords. Accords are broad olfactory descriptors (e.g., woody, floral, aromatic, and gourmand) that summarize a fragrance’s dominant scent characteristics and constitute the standard descriptive vocabulary used by the fragrance industry and consumers (Edwards, 2020). For 67% of the fragrances, we also observe a sample of user reviews. Ratings and reviews are generated by a self-selected user population that is likely more engaged than the average consumer. See Web Appendix A for additional details on the dataset.

Consumer reviews have become an important source of behavioral data in marketing because they provide rich, unsolicited accounts of how consumers perceive, evaluate, and experience products (Boughanmi, Jedidi, and Jedidi, 2025; Lee and Bradlow, 2011; Tirunillai and Tellis, 2014). Their growing availability has also stimulated advances in marketing text analysis, enabling researchers to extract behavioral insights from large-scale unstructured text (Berger et al., 2020; Büschken and Allenby, 2016). Unlike structured product attributes or numerical ratings, reviews reveal the language consumers use to explain their evaluations and describe product characteristics. In our setting, they complement the structured fragrance data by revealing how consumers interpret different forms of directional atypicality, providing a behavioral explanation for the directional patterns identified in the structured-data analysis.

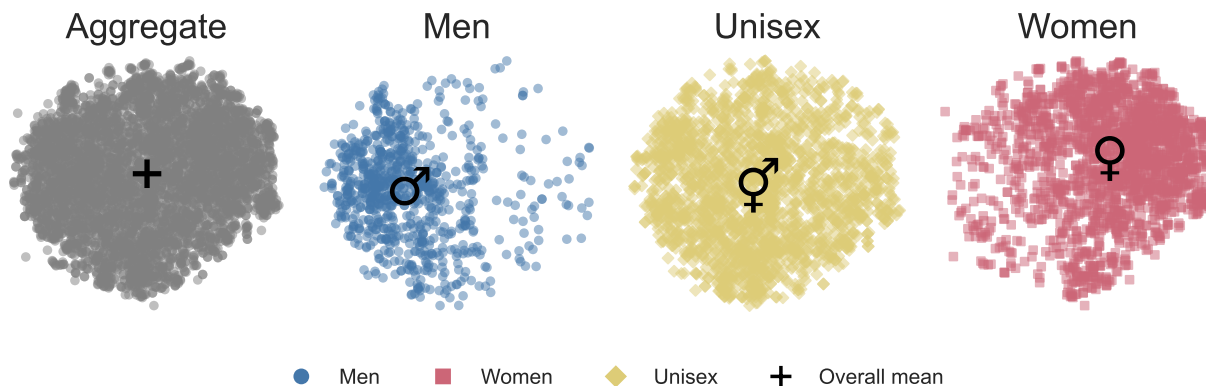
2.1 Descriptive Statistics

Ratings range from 1 to 5 and capture overall perceived appeal. The median rating is 4 (SD = 0.517), indicating moderate dispersion. Mean ratings vary slightly by target segment: men’s 3.979 (SD = 0.509), women’s 3.948 (SD = 0.465), and unisex 4.006 (SD = 0.568). Although individual ratings are unavailable, we observe aggregate counts, with the median fragrance having 27 ratings.

Each fragrance’s scent profile is described using up to 90 olfactory accords (i.e., composite scent notes such as woody, floral, or citrus). Overall, woody (66%), powdery (51%), and citrus (49%) accords are the most common. Men’s fragrances are dominated by woody (85%), aromatic (78%), and fresh spicy (67%) accords, whereas women’s fragrances emphasize sweet (61%), floral (59%), and powdery (59%) accords. Unisex fragrances lie between these profiles, combining accords prevalent in both segments. Their most common accords are woody (67%), powdery (47%), and warm spicy (46%), reflecting an olfactory composition designed to appeal across gender categories. The distribution of these accords across fragrances is reported in Web Appendix A, both by segment and overall.

Figure 1 presents a two-dimensional t-distributed stochastic neighbor embedding (t-SNE; [Van der Maaten and Hinton, 2008](#)) projection of 5,000 randomly sampled fragrances based on their olfactory profiles (accords coded as binary indicators), with points colored by segment. The aggregate panel shows a dense cloud that obscures the underlying segment structure. In contrast, segment-level panels reveal clear positioning differences: men’s fragrances cluster to the left, women’s to the right, and unisex fragrances in between. Within each segment, fragrances closer to the centroid are more typical, whereas those farther away are more atypical. We leverage this segment structure to identify where differentiation is rewarded, uncovering patterns not visible in the aggregate distribution.

Figure 1: t-SNE Visualization of Fragrance Accord Profiles



Note. The t-SNE embedding is based on fragrances’ accord profiles using a random sample of 5,000 fragrances from the full dataset. Each point represents a fragrance, and symbols denote segment centroids. The left panel shows the aggregate distribution, while the right panels display the men’s, unisex, and women’s segments separately. Although the aggregate distribution appears as a single undifferentiated cloud, the segment-level plots reveal distinct structures, with unisex fragrances occupying an intermediate position between men’s and women’s fragrances. This contrast illustrates how aggregation masks underlying segment-level structure.

2.2 Atypicality Distributions

Because men’s, women’s, and unisex fragrances follow systematically different scent norms, a single global benchmark would blur these differences. In the categorization literature, typicality is defined relative to segment prototypes rather than in absolute terms (Loken, 2006; Rosch, 1978). Similarly, work on category spanning shows that evaluations depend on how products relate to category-specific expectations (Hsu, 2006; Pontikes, 2012; Zuckerman, 1999). Thus, atypicality is inherently category-specific, as deviations are interpreted relative to the norms of the relevant reference set. We therefore compare each fragrance to the fragrances that define its segment.

We measure fragrance atypicality as the Jaccard distance (Jaccard, 1912) between a fragrance’s accord composition and those of other fragrances within its segment. This measure is well suited to our binary data, which indicate the presence or absence of accords in each fragrance’s olfactory profile (Toubia and Netzer, 2017). For robustness, we also consider a range of alternative measures, including pairwise distances (Hamming, cosine, and Euclidean), model-based measures (negative log-likelihood), prototype-based measures (cosine and Euclidean distance to segment centroids), and local neighborhood measures based on k -nearest neighbors ($k = 500$ and $k = 1,000$). Results are qualitatively consistent across all specifications and are reported in Web Appendix B.

Within each segment, fragrance i is represented by the set of accords it contains, denoted A_i . For two fragrances i and j , the Jaccard similarity is defined as the ratio of the size of the intersection to the union of their accord sets:

$$J(i, j) = \frac{|A_i \cap A_j|}{|A_i \cup A_j|}.$$

The corresponding Jaccard distance is $d_J(i, j) = 1 - J(i, j)$. We define atypicality as the average Jaccard distance between fragrance i and all other fragrances in its target segment:

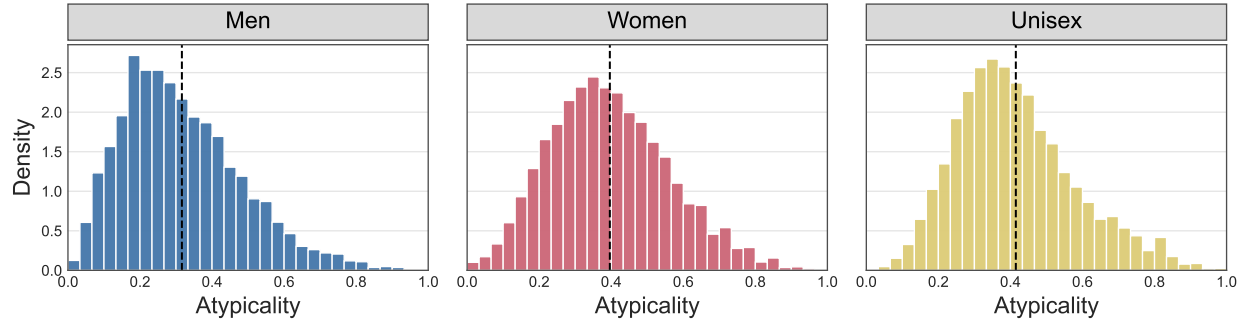
$$\text{Atypicality}_i = \frac{1}{N_s - 1} \sum_{\substack{j=1 \\ j \neq i}}^{N_s} d_J(i, j),$$

where N_s is the number of fragrances in the segment. Higher values indicate that a fragrance’s accord composition deviates more from others in the same segment. The measure is normalized to lie between 0 and 1 within each segment.

Figure 2 shows the distribution of atypicality across segments. Mean atypicality differs slightly across men (mean = 0.316, SD = 0.165), women (mean = 0.396, SD = 0.166), and unisex (mean = 0.415, SD = 0.165) fragrances. A one-way ANOVA indicates that these differences are statistically significant ($F(2, 61,865) = 1,380.568$, $p < 0.001$), but modest in magnitude ($\eta^2 = 0.043$).

To illustrate how atypicality varies within segments, we consider representative examples, reporting atypicality values in parentheses. Among women’s fragrances, the iconic *Chanel No. 5* (0.436) is more atypical than *J’Adore* by Dior (0.273), reflecting a more unconventional blend of aldehydic and powdery notes relative to a more standard floral profile. Among men’s fragrances, *1 Million* by Paco Rabanne (0.448) is more atypical than *Sauvage* by Dior (0.097), reflecting a richer spicy-leather composition relative to a more conventional fresh citrus-woody structure. Among

Figure 2: Distribution of Olfactory Atypicality Across Gender Segments



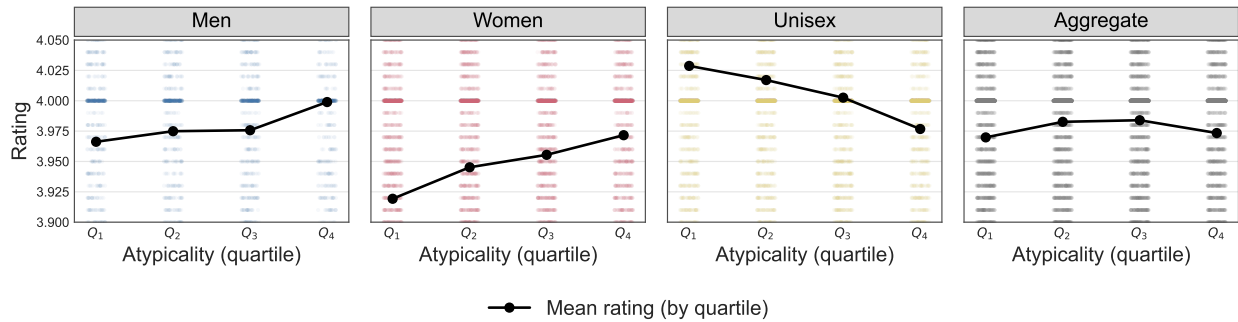
Note. The figure shows the distribution of atypicality scores for men's, women's, and unisex fragrances. Atypicality is defined as the average Jaccard distance between a fragrance and all other fragrances in its target segment, with higher values indicating greater deviation from segment norms. Dashed vertical lines denote segment-specific means.

unisex fragrances, *Fleur de Peau* by Diptyque (0.431) departs more from segment norms than *Baccarat Rouge 540* by Maison Francis Kurkdjian (0.323), reflecting a more pronounced shift away from a typical amber-woody profile.

2.3 Atypicality and Consumer Ratings: Descriptive Evidence

Figure 3 plots average ratings by atypicality quartile for men's, women's, and unisex fragrances. Ratings increase with atypicality for men and women but decline for unisex, where the highest ratings occur in the lowest quartile. Thus, higher atypicality is associated with higher ratings in the men's and women's segments, whereas the opposite pattern is observed in the unisex segment. In contrast, the aggregate pattern (right panel) appears non-monotonic, reflecting aggregation across segments with opposing relationships and obscuring the segment-level patterns.

Figure 3: Atypicality-Ratings Relationship Across Segments and Aggregate



Note. The figure displays mean ratings across atypicality quartiles for men's, women's, and unisex fragrances, as well as for the aggregate sample. Atypicality is defined as the distance from the segment prototype within segments and from the full set of fragrances in the aggregate analysis. Ratings increase with atypicality in men's and women's segments, but decrease in the unisex segment, indicating systematic heterogeneity across segments. In contrast, the aggregate pattern is non-monotonic, with ratings peaking at intermediate levels of atypicality.

3 Empirical Analysis

We examine whether a fragrance’s atypicality is systematically related to consumer evaluations and whether this relationship depends on category structure. We first analyze the relationship between atypicality and consumer ratings at the segment and aggregate levels. We then examine category structure and introduce measures of directional atypicality to assess whether the value of differentiation depends on the direction, rather than simply the magnitude, of deviation from category norms.

3.1 Relationship Between Atypicality and Consumer Evaluations

We begin by estimating linear and nonlinear models relating atypicality to consumer evaluations within each fragrance segment. We then compare these results with aggregate estimates based on pooled data to assess whether segment-specific relationships are obscured when atypicality is measured relative to a common benchmark.

3.1.1 Baseline Linear Model

We estimate the following baseline specification:

$$\text{Rating}_i = \alpha_{B(i)} + \theta_{D(i)} + \beta \text{Atypicality}_i + \gamma^\top \mathbf{z}_i + \varepsilon_i, \quad (1)$$

where Rating_i is the average rating of fragrance i , and Atypicality_i is the average Jaccard distance between fragrance i and other fragrances in its segment. Higher values indicate greater dissimilarity from segment peers. The coefficient β captures the association between such deviations and consumer evaluations.

The specification includes several layers of controls. Brand fixed effects, $\alpha_{B(i)}$, account for persistent differences in reputation and positioning, while decade fixed effects, $\theta_{D(i)}$, capture time-varying trends in fragrance design and preferences. The vector $\mathbf{z}_i$ includes observable product characteristics, including *nez* (perfumer) indicators, which control for differences in creative style, expertise, and resources, as well as product-level controls (e.g., the number of accords and the number of ratings) that capture structural complexity and exposure, both of which may affect atypicality and perceived quality. We do not include accord-level controls because atypicality is constructed from these features; including them would largely partial out the variation of interest, raising a “bad control” concern (Angrist and Pischke, 2009) and inducing severe multicollinearity (see Web Appendix C). Finally, standard errors are clustered at the brand level to account for potential correlation in unobserved factors among fragrances produced by the same brand, which could otherwise lead to underestimated standard errors and overstated statistical significance.

The analysis leverages within-brand and within-period variation in atypicality, allowing us to examine its association with ratings while accounting for systematic differences in brand quality and temporal trends. It also captures heterogeneity across segments by estimating the model separately for men’s, women’s, and unisex fragrances.

Table 1 reports the estimated relationship between atypicality and consumer ratings across

segments. The models explain a meaningful amount of variation, with R^2 values of 0.206 (men), 0.166 (women), and 0.170 (unisex).

A clear asymmetry emerges across segments. For men’s and women’s fragrances, atypicality is positively associated with ratings (men: 0.124, $p = 0.026$; women: 0.116, $p < 0.001$), corresponding to an increase of roughly 0.12 rating points per unit of atypicality. In contrast, the effect for unisex fragrances is negative and significant (-0.059 , $p = 0.040$), indicating that greater atypicality lowers ratings. The positive effects in gendered segments are roughly twice as large as the negative effect for unisex fragrances, highlighting asymmetry in both direction and magnitude.

Table 1: Effects of Atypicality on Consumer Ratings

Variable	Men			Women			Unisex			Aggregate		
	Est.	SE	p	Est.	SE	p	Est.	SE	p	Est.	SE	p
<i>Panel A: Linear specification</i>												
Atypicality	0.124	0.056	0.026	0.116	0.029	< 0.001	-0.059	0.029	0.040	0.042	0.018	0.019
<i>Panel B: Nonlinear specification</i>												
Q_1 (Most typical)	—			—			—			—		
Q_2	0.016	0.013	0.241	0.013	0.009	0.156	-0.005	0.007	0.509	0.012	0.005	0.030
Q_3	0.014	0.014	0.298	0.018	0.009	0.052	-0.014	0.010	0.158	0.010	0.005	0.057
Q_4 (Most atypical)	0.057	0.022	0.008	0.042	0.013	0.001	-0.028	0.011	0.011	0.017	0.007	0.012
R^2		0.206			0.166			0.170			0.146	
Adjusted R^2		0.136			0.131			0.134			0.128	
Observations		10,786			25,468			25,614			61,868	
Controls												
Brand FE		Yes			Yes			Yes			Yes	
Decade FE		Yes			Yes			Yes			Yes	
Perfumer		Yes			Yes			Yes			Yes	
Rating-count		Yes			Yes			Yes			Yes	
Accord-count		Yes			Yes			Yes			Yes	
Segment		No			No			No			Yes	
Clustered SEs		Brand			Brand			Brand			Brand	

Note. The dependent variable is average consumer rating. Entries report coefficient estimates, brand-clustered standard errors, and exact p -values. The quartile specification omits Q_1 (most typical). All models include brand and decade fixed effects, perfumer controls, rating-count controls, and accord-count controls. The aggregate model additionally controls for target segment.

3.1.2 Nonlinear Specification

We examine nonlinearities in the relationship between atypicality and ratings. Prior work suggests that the effect of deviation from segment norms is often non-monotonic, with moderate deviation typically preferred (e.g., [Askin and Mauskopf, 2017](#); [Brewer, 1991](#); [Meyers-Levy and Tybout, 1989](#)). To capture such patterns, we partition fragrances into atypicality quartiles.¹ Results are robust to alternative discretizations (i.e., terciles and quintiles; [Web Appendix D](#)).

We replace the continuous atypicality measure in Eq. (1) with quartile indicators and estimate

¹We do not use a quadratic specification due to the high correlation between atypicality and its square (0.959 for men, 0.969 for women, 0.974 for unisex, and 0.969 in the aggregate), resulting in severe multicollinearity.

the following specification:

$$\text{Rating}_i = \alpha_{B(i)} + \theta_{D(i)} + \sum_{q=2}^4 \beta_q \mathbf{1}\{\text{Atypicality}_i \in Q_q\} + \boldsymbol{\gamma}^\top \mathbf{z}_i + \varepsilon_i, \quad (2)$$

where Q_q denotes the q -th atypicality quartile, with Q_1 as the reference baseline.

Table 1 shows that the effects of atypicality are concentrated in the upper tail of the distribution. For men’s and women’s fragrances, coefficients for Q_2 and Q_3 are small and generally insignificant, indicating limited differences relative to the most typical fragrances. By contrast, the most atypical fragrances (Q_4) receive significantly higher ratings, with coefficients of 0.057 (men) and 0.042 (women), both statistically significant at $p < 0.05$. The pattern reverses for unisex fragrances. While Q_2 and Q_3 show little difference relative to Q_1 , the most atypical fragrances (Q_4) receive significantly lower ratings (-0.028 , $p = 0.011$). Overall, the relationship between atypicality and evaluation is nonlinear and concentrated in the upper tail, with extreme atypicality rewarded in gendered segments but penalized in the unisex segment, underscoring the importance of segment structure in shaping the returns to differentiation.

Table 2: Substantive Magnitude of Estimated Rating Differences

Segment	Direction	Rating Shift	Percentile-Rank Shift	SD Units	IQR Units
Men	$Q_1 \rightarrow Q_4$	0.057	12.4	0.112	0.121
Women	$Q_1 \rightarrow Q_4$	0.042	5.4	0.090	0.095
Unisex	$Q_4 \rightarrow Q_1$	0.028	9.9	0.050	0.053

Note. Rating shifts are predicted differences in average fragrance ratings. Percentile-rank shifts are computed relative to the empirical distribution of average fragrance ratings within each segment. For men and women, the reported direction is from the most typical quartile (Q_1) to the most atypical quartile (Q_4), reflecting the higher ratings of atypical fragrances in gendered segments. For unisex fragrances, the reported direction is reversed because typical fragrances receive higher ratings; equivalently, the raw Q_4 versus Q_1 coefficient for unisex fragrances is -0.028 .

Although the estimated coefficients are modest in absolute terms, this is expected because the dependent variable is an aggregate fragrance-level rating with limited dispersion. To assess substantive magnitude, Table 2 translates the estimated rating differences into percentile-rank shifts in the empirical distribution of average fragrance ratings, as well as standardized changes relative to the segment-specific standard deviation and interquartile range.

For men’s fragrances, moving from the most typical quartile (Q_1) to the most atypical quartile (Q_4) is associated with a 0.057-point increase in predicted rating, equivalent to a 12.4 percentile-rank shift. For women’s fragrances, the corresponding movement from Q_1 to Q_4 is associated with a 0.042-point increase, equivalent to a 5.4 percentile-rank shift. In contrast, for unisex fragrances, where typical products are rated more highly, moving from the most atypical quartile (Q_4) to the most typical quartile (Q_1) is associated with a 0.028-point improvement, equivalent to a 9.9 percentile-rank shift. These effects also correspond to approximately 0.05–0.11 standard deviations and 0.05–0.12 interquartile ranges across segments. Thus, although the raw rating differences are small on a five-point scale, they represent substantively meaningful changes in products’ relative standing within their segments.

3.1.3 Aggregate Patterns in Atypicality and Ratings

We next examine how the segment-level results align with those from an aggregate analysis of the data. To do so, we follow [Askin and Mauskopf \(2017\)](#) and re-estimate the relationship between atypicality and ratings at the market level. Consistent with their approach, atypicality is measured as a weighted average of pairwise distances between a fragrance and others, such that distances to fragrances in the same segment receive greater weight, while distances to more distant segments receive lower weight. Web Appendix E provides computational details.

Table 1 reports aggregate results based on Jaccard distance; results using cosine distance are qualitatively similar and reported in Web Appendix E. The linear effect of atypicality is positive but modest (0.042 , $p = 0.019$), masking heterogeneity across segments. The quartile specification shows a generally positive relationship between atypicality and ratings: relative to the most typical fragrances (Q_1), ratings are higher for Q_2 (0.012 , $p = 0.030$), Q_3 (0.010 , $p = 0.057$), and Q_4 (0.017 , $p = 0.012$). While the aggregate specification suggests that more atypical fragrances receive higher evaluations, it obscures substantial differences across segments.

These aggregate results underscore the importance of defining atypicality relative to an appropriate category prototype ([Barsalou, 1985](#); [Loken, 2006](#); [Rosch, 1978](#)). In the aggregate analysis, atypicality is measured relative to a common reference that pools fragrances from different segments. In our setting, however, fragrances are organized into distinct segments with their own local prototypes. As a result, the pooled relationship between atypicality and ratings is ambiguous, combining segment-specific patterns without representing any single one. In particular, positive returns to atypicality in men’s and women’s fragrances and negative returns in unisex fragrances yield an aggregate pattern that obscures the underlying evaluative structure. These findings suggest that the value of atypicality is inherently reference-dependent and cannot be understood without accounting for segment-level heterogeneity.

3.2 Category Structure and Directional Atypicality

The preceding results show that the relationship between atypicality and ratings differs across segments. This asymmetry suggests that not all forms of atypicality are equally valued and points to the importance of where fragrances shift in segment space. We further characterize these patterns through directional atypicality, using both structured fragrance accord data and unstructured consumer reviews. Because directional atypicality measures are derived from the same underlying accord structure as atypicality itself, they should be interpreted as descriptive decompositions of atypicality rather than causal mediators.

To characterize directional atypicality, we estimate a multinomial logit (MNL) model that maps fragrances into segment space and uses predicted segment probabilities to capture both the magnitude and direction of deviation (e.g., a higher unisex probability for a men’s fragrance indicates greater proximity and directional shift toward the unisex segment). We then examine whether these directional measures help explain variation in the relationship between atypicality and ratings. If the probability coefficients are small or insignificant, ratings are not systematically associated with this model-implied direction of movement. If the atypicality coefficient attenuates once

directional measures are included, this suggests that evaluations are more strongly associated with directional shifts than with atypicality alone. Finally, we visualize these patterns using geometric representations.

Importantly, these predicted probabilities are not measures of consumers’ perceived category membership. Because the MNL is trained on manufacturer-assigned segment labels and accord profiles, they capture model-implied proximity to labeled segment prototypes in accord space. Thus, movement toward the unisex segment reflects greater accord-based resemblance to fragrances labeled as unisex, rather than consumers’ subjective classification of the fragrance.

3.2.1 Directional Atypicality: Evidence from Structured Accord Data

We estimate an MNL model that specifies a fragrance’s segment (men’s, women’s, or unisex) as the dependent variable and olfactory accords as predictors. While atypicality captures the magnitude of deviation, predicted segment probabilities capture directional shifts and proximity by locating each fragrance relative to segment prototypes, with higher probabilities indicating greater similarity to a segment’s prototype. The model fits the data well ($\chi^2 = 33,758$, $p < 0.001$, pseudo- $R^2 = 0.264$). Estimated coefficients (Web Appendix F) are used to compute predicted segment probabilities.

We now examine whether controlling for directional atypicality alters the relationship between atypicality and ratings. To test this, we augment the baseline regressions with the log-odds of predicted segment probabilities:²

$$\text{Rating}_i = \alpha_{B(i)} + \theta_{D(i)} + \beta \text{Atypicality}_i + \sum_{s \in \mathcal{S}_{-g}} \lambda_s \log \left(\frac{\hat{P}_{is}}{\hat{P}_{ig}} \right) + \gamma^\top \mathbf{z}_i + \varepsilon_i, \quad (3)$$

where $\hat{P}_{is}$ is the predicted probability that fragrance i belongs to segment s , and $\hat{P}_{ig}$ is the probability of its own segment g . The set $\mathcal{S}_{-g}$ includes all segments other than g , so each term measures the model-implied affinity of fragrance i to segment s relative to its assigned segment g . These log-odds capture directional atypicality by indicating both the direction and magnitude of deviation in segment space. Accordingly, for men’s (women’s) fragrances, they capture shifts toward women and unisex (men and unisex), and for unisex fragrances, shifts toward men and women. The λ_s parameters capture the associations between ratings and directional movement in accord-defined segment space.

Table 3 shows that the atypicality coefficient becomes insignificant for men’s and women’s fragrances once we control for the log-odds of predicted segment probabilities (men: $\beta = 0.019$, $p = 0.760$; women: $\beta = -0.053$, $p = 0.105$), but remains negative and significant for unisex fragrances ($\beta = -0.132$, $p < 0.001$). In the gendered segments, ratings vary systematically with directional shifts in segment membership: shifts toward unisex (relative to the own segment) increase ratings, whereas shifts toward the other gendered segment decrease them (men: $\lambda = 0.040$

²We use log-odds relative to the fragrance’s own segment g because predicted segment probabilities are compositional (sum to one) and thus lie on a constrained simplex; log-ratio transformations provide appropriate, scale-invariant coordinates (Aitchison, 1982). This also avoids singularity when all probabilities are included directly. All predicted probabilities are bounded away from zero, ensuring well-defined log-ratios.

Table 3: Effects of Atypicality and Predicted Segment Affinity on Consumer Ratings

Variable	Men			Women			Unisex		
	Est.	SE	<i>p</i>	Est.	SE	<i>p</i>	Est.	SE	<i>p</i>
Atypicality	0.019	0.062	0.760	-0.053	0.032	0.105	-0.132	0.035	< 0.001
<i>Segment affinity (log-odds ratios)</i>									
Women vs. Men	-0.014	0.005	0.006						
Unisex vs. Men	0.040	0.009	< 0.001						
Men vs. Women				-0.020	0.003	< 0.001			
Unisex vs. Women				0.054	0.006	< 0.001			
Men vs. Unisex							-0.014	0.003	< 0.001
Women vs. Unisex							-0.015	0.002	< 0.001
R^2		0.208			0.168			0.172	
Adjusted R^2		0.138			0.133			0.136	
Observations		10,786			25,468			25,614	
Controls									
Brand FE		Yes			Yes			Yes	
Decade FE		Yes			Yes			Yes	
Perfumer		Yes			Yes			Yes	
Rating-count		Yes			Yes			Yes	
Accord-count		Yes			Yes			Yes	
Clustered SEs		Brand			Brand			Brand	

Note. The dependent variable is consumer rating. Entries report coefficient estimates, brand-clustered standard errors, and exact *p*-values. All models include brand and decade fixed effects, perfumer controls, rating-count controls, and accord-count controls.

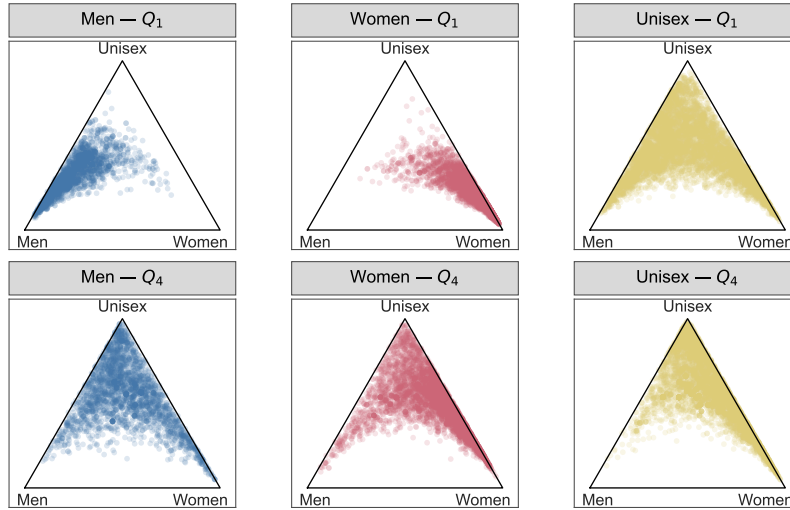
vs. -0.014 ; women: $\lambda = 0.054$ vs. -0.020 , all $p < 0.01$). In contrast, for unisex fragrances, shifts toward either gendered segment are penalized (men: $\lambda = -0.014$; women: $\lambda = -0.015$, both $p < 0.001$). These results indicate that the relationship between atypicality and ratings is strongly associated with directional shifts in segment space: movement toward the shared unisex region is rewarded for men’s and women’s fragrances, whereas for unisex fragrances, deviations from the prototype are penalized.

Regressions of atypicality on the corresponding log-odds yield only moderate fit across segments (average $R^2 = 0.365$), indicating that directional atypicality captures variation distinct from atypicality.

Shifts in predicted segment probabilities: Figure 4 visualizes these classification patterns using a simplex representation based on the predicted segment probabilities. Men’s and women’s fragrances exhibit directional shifts toward the unisex region as atypicality increases, reflected in a higher probability of being classified as unisex. In contrast, unisex fragrances remain concentrated in the unisex region, but deviations from this core position are asymmetric, shifting more toward the women’s segment than the men’s segment. These patterns further illustrate directional atypicality, with movements in segment space following systematic rather than diffuse directions.

Movements in accord space: Figure 5 provides a geometric synthesis of the underlying mechanism using linear discriminant analysis (LDA), which, like the MNL model, uses accord features to distinguish fragrance segments. The LDA is estimated using only the accords identified as signifi-

Figure 4: Simplex of Predicted Segment Probabilities by Atypicality Quartile (Q_1 vs. Q_4)



Note. The figure plots predicted probabilities of segment membership (men’s, women’s, and unisex) in a simplex representation for fragrances in the lowest (Q_1) and highest (Q_4) atypicality quartiles. From Q_1 to Q_4 , men’s and women’s fragrances shift toward the unisex region. In contrast, unisex fragrances remain concentrated near the unisex region, with a slight shift toward the women’s region at Q_4 .

cant in the MNL model, omitting insignificant features to focus on the dimensions driving segment differentiation. The resulting discriminant functions separate segments clearly (Wilks’ $\Lambda = 0.558$, $p < 0.001$), confirming that the two-dimensional representation captures meaningful differences in accord composition. The first discriminant dimension primarily reflects gender orientation, separating traditionally masculine accords on the left (e.g., oud, tobacco, leather, whiskey, aromatic, and woody notes) from traditionally feminine accords on the right (e.g., floral, white floral, rose, and fruity notes). The second discriminant dimension reflects accord prevalence, with relatively uncommon or niche accords located toward the top (e.g., oily, oud, wine, cannabis, metallic, and beeswax) and more commonly used accords toward the bottom (e.g., lavender, citrus, aquatic, marine, and fruity). The accords contributing most strongly to each discriminant dimension are reported in Web Appendix G.

Figure 5 also plots the mean discriminant scores for fragrances in the lowest (Q_1) and highest (Q_4) atypicality quartiles within each segment. Because all fragrances are described using the same industry-standard accord vocabulary, movements in this common category space correspond to systematic changes in their olfactory profiles and, in turn, in their similarity to the masculine, feminine, and unisex prototypes. As atypicality increases, men’s and women’s fragrances move toward the central unisex region, whereas unisex fragrances remain near the center, exhibiting only a slight shift toward the women’s region. These patterns are consistent with the preceding analyses, indicating that highly rated gendered fragrances move toward a shared cross-segment region, whereas highly rated unisex fragrances remain anchored near a balanced central prototype. At this stage, the analysis identifies where fragrances move within category space but not the semantic characteristics underlying those movements. We address this question in the next section by examining how consumers describe fragrances exhibiting different forms of directional atypicality.

In sum, the value of atypicality depends not only on how far a product deviates, but on where it moves relative to the relevant segment prototype. Among men’s and women’s fragrances, higher evaluations are associated with atypical products that shift toward a shared, interpretable region, whereas among unisex fragrances, atypical products receive lower evaluations when they shift away from a balanced prototype.

This directional perspective highlights that evaluation is reference-dependent: in segmented markets with multiple local prototypes, there is no single global optimum of atypicality. Instead, pooled patterns reflect a mixture of segment-specific relationships that can obscure the underlying evaluative structure. More broadly, successful differentiation requires deviations that remain interpretable relative to the appropriate prototype.

4 Directional Atypicality: Evidence from Consumer Reviews

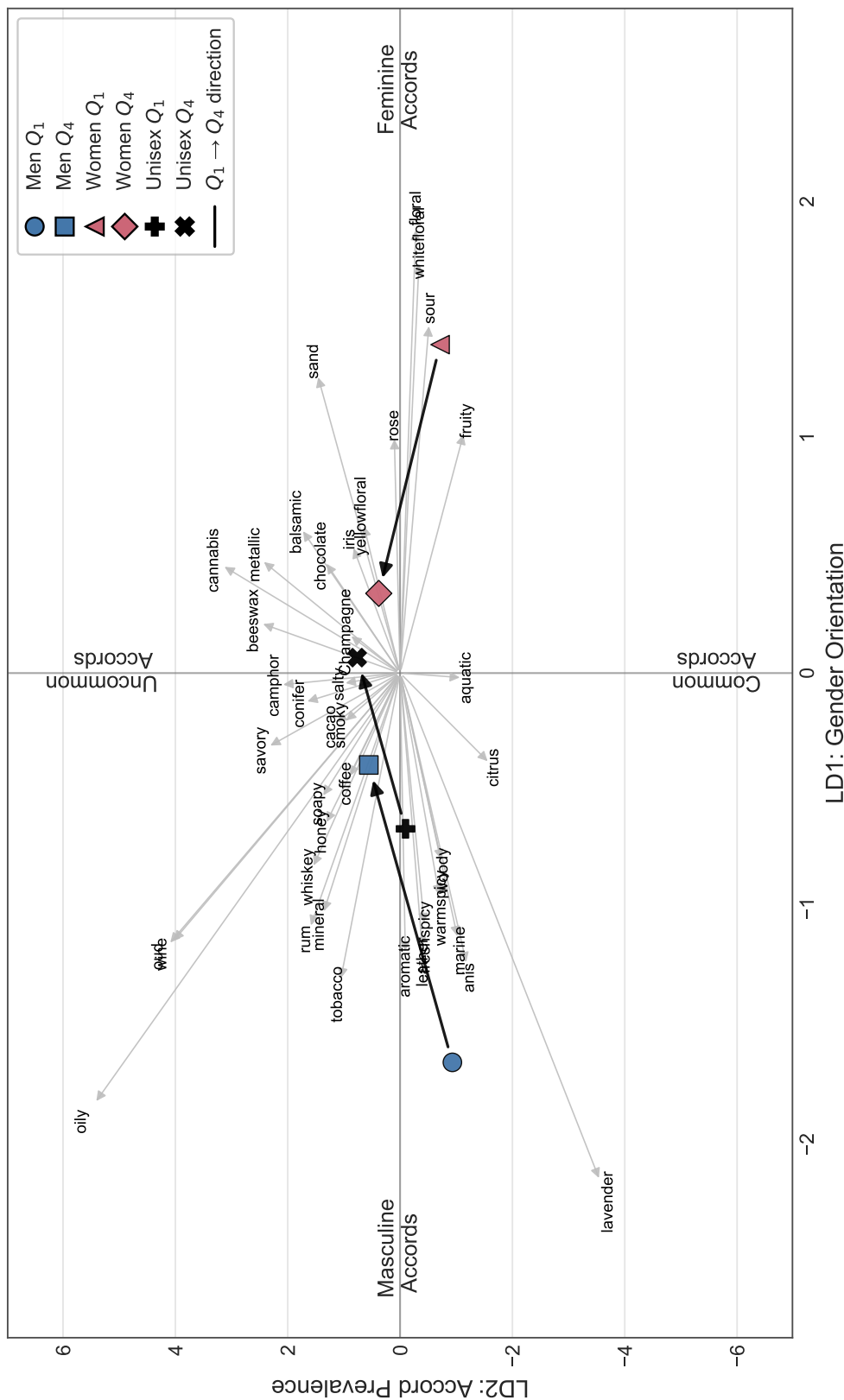
The preceding analyses establish how fragrances move in category space and how these movements relate to consumer evaluations, but they do not explain how consumers interpret them. To address this question, we augment our dataset with textual reviews from *Fragrantica*, covering 67% of the fragrances in our sample and totaling 1,017,974 reviews (approximately 24 reviews per fragrance). Because fragrances with reviews differ systematically from the underlying fragrance population, we first construct a balanced fragrance sample using a stratified propensity-based procedure within each gender segment. The resulting sample contains 4,500 men’s fragrances, 9,600 women’s fragrances, and 7,100 unisex fragrances (see Web Appendix H for details of the sample construction and balance diagnostics). We then retain fragrances in the lowest and highest atypicality quartiles (Q_1 and Q_4), yielding 52,548, 93,336, and 45,869 reviews for the men’s, women’s, and unisex segments, respectively. These reviews are randomly partitioned into two equal, non-overlapping samples: a discovery sample used to identify semantic themes and a validation sample reserved for confirmatory analyses.

We proceed in two phases. In the exploratory phase, we first embed sentences from reviews in the discovery sample using sentence-transformer embeddings (Xiao et al., 2023). We then estimate segment-specific random forest models to identify the most discriminative sentences for the focal fragrance group (Q_4 for men’s and women’s fragrances; Q_1 for unisex fragrances) and cluster their embeddings into semantic themes. In the confirmatory phase, we assign each sentence in the validation sample of reviews to the theme of its nearest neighbor in the exploratory sample using embedding-based matching and examine whether theme prevalence differs systematically across atypicality quartiles.

4.1 Exploratory Phase

We conduct the exploratory analysis at the sentence level to capture distinct evaluative expressions. Following sentence-based approaches to analyzing customer reviews (Büschken and Allenby, 2016), reviews in the discovery sample from the Q_1 and Q_4 quartiles are first split into individual sentences and represented using normalized 1024-dimensional sentence embeddings (Xiao et al., 2023). Sentence embeddings capture semantic similarity beyond shared keywords, allowing conceptually

Figure 5: LDA Map of Q_1 vs. Q_4 Fragrances Across Segments Using Accord Features



Note. Vectors denote accords, and points represent the mean discriminant scores of Q_1 and Q_4 fragrances within each segment. The horizontal axis reflects gender orientation, and the vertical axis reflects accord prevalence, with higher values corresponding to less common accords. As atypicality increases, men's and women's fragrances move toward the central unisex region, whereas unisex fragrances remain near the center, exhibiting only a slight shift toward the women's region.

related expressions to be represented nearby in the embedding space. This yields 129,676, 236,303, and 116,868 sentences for the men’s, women’s, and unisex segments, respectively. We then estimate segment-specific random forest models to identify the sentences most predictive of membership in the focal fragrance group (Q_4 for men’s and women’s fragrances; Q_1 for unisex fragrances).³ The resulting classifiers achieve out-of-sample AUCs of 0.623, 0.613, and 0.635 for the men’s, women’s, and unisex segments, respectively, indicating that sentence-level semantic content contains meaningful information for distinguishing the focal fragrance groups from their comparison groups.

To identify semantic themes that distinguish the focal fragrance groups, we follow recent marketing research using consumer-generated text to extract behavioral insights (Berger et al., 2020). Within each segment, we retain the top 10% of sentences ranked by their predicted probability of belonging to the focal group (Q_4 for men’s and women’s fragrances; Q_1 for unisex fragrances). Focusing on this most discriminative subset allows the analysis to identify the semantic patterns that most clearly distinguish the focal group within each segment. This yields 12,968, 23,631, and 11,687 highly discriminative sentences for the men’s, women’s, and unisex segments, respectively. We then cluster the corresponding sentence embeddings using k -means to identify semantically coherent themes. Based on silhouette criteria, we identify 13 themes for men’s fragrances, 17 themes for women’s fragrances, and 15 themes for unisex fragrances. Table 4 summarizes the relative frequency of the discovered themes among these retained sentences. Additional details of the theme discovery procedure are provided in Web Appendix I, and representative sentences for each theme are reported in Web Appendix J.

For men’s and women’s fragrances, the themes reveal several recurring patterns of differentiation. One involves olfactory recombination through the incorporation or intensification of particular accords. In men’s fragrances, atypicality is associated with themes centered on oud, leather, and tobacco (e.g., “there’s some oud in the mix,” “leather, leather and a bit more leather,” and “giving the tobacco a sweet and smoky vibe”; M2, M4, M12), together with cross-gender recombination through sweet and gourmand elements that soften conventional masculine profiles (M7). In women’s fragrances, atypicality is associated both with the intensification of traditionally feminine accords, including rose, tuberose, white florals, creamy gourmand, and coconut notes, and with cross-gender recombination through spicy, woody, and resinous accords that overlap with traditionally masculine olfactory profiles (e.g., “this is THE rose perfume,” “a beautifully blended scent with a focus on tuberose,” “this smells like a creamy coconut with sugar,” and “the scent is spicy, sweet, and woody”; W2, W4–W6, W10, W13). These themes suggest that atypical fragrances are characterized by distinctive combinations of recognizable olfactory materials, either by intensifying established category accords or by recombining elements across traditional gender boundaries.

Another recurring pattern concerns evaluative, sensory, and experiential impressions. Both men’s and women’s fragrances contain themes emphasizing strong positive evaluations, hedonic richness, and perceptions of artistry and refinement. Representative expressions include “this is a spectacular scent,” “this stuff is gorgeous,” “the artistry is exquisite,” and “this fragrance is a wow-zer” (M3, M6, W11, W12, W14). Women’s fragrances also contain themes highlighting atmospheric complexity and novelty, with reviewers describing fragrances as “interesting,” “lush,”

³We use random forests to capture nonlinearities and interaction effects in high-dimensional text embeddings.

Table 4: Distribution of Discovered Themes Across Exploratory and Confirmatory Samples

Segment	Theme	Exploratory (%)	Confirmatory (%)
Men	M1: Seasonal Suitability and Cold-Weather Appeal	2.46	2.67
	M2: Material-Centric Recombination (Oud)	4.50	4.25
	M3: Intense Positive Evaluation and Aesthetic Appeal	11.37	15.08
	M4: Material-Centric Recombination (Leather)	7.42	7.79
	M5: Ownership, Sampling, and Acquisition Experiences	7.77	7.98
	M6: Hedonic Richness and Emotional Appeal	7.73	7.09
	M7: Cross-Gender Recombination (Sweet and Gourmand Accords)	9.08	8.86
	M8: Intensity and Sensory Presence	9.77	9.18
	M9: Idiosyncratic Olfactory Impressions and Sensory Ambiguity	10.02	10.10
	M10: Negative Expectation Disconfirmation and Taste-Based Rejection	9.47	8.18
	M11: Gender Boundary Blurring (Unisex Perception)	3.45	4.02
	M12: Within-Category Recombination (Tobacco and Smoky Accords)	9.24	7.55
	M13: Ambivalence and Taste-Based Uncertainty	7.72	7.26
	Nb. of Sentences	12,968	14,025
Women	W1: Evocative Depth and Atmospheric Complexity	6.99	6.27
	W2: Within-Category Recombination (Rose Intensification)	3.54	3.37
	W3: Negative Expectation Disconfirmation (Category Ambiguity)	5.34	4.66
	W4: Within-Category Recombination (Creamy Gourmand Accords)	8.57	8.29
	W5: Within-Category Recombination (Coconut Gourmand Accords)	3.24	3.28
	W6: Cross-Gender Recombination (Spicy and Resinous Accords)	8.30	6.95
	W7: Novelty and Distinctiveness	8.95	11.37
	W8: Performance Intensity and Wearability	3.44	4.12
	W9: Gender Boundary Blurring (Unisex Perception)	1.17	1.86
	W10: Within-Category Recombination (Floral Intensification)	3.71	4.16
	W11: Perceived Artistry and Refined Appeal	6.13	5.92
	W12: Intense Positive Affect and Enthusiastic Evaluation	6.48	6.90
	W13: Within-Category Recombination (White Floral Intensification)	5.61	5.34
	W14: Evocative Sensory Appeal and Hedonic Richness	10.08	11.67
	W15: Accord-Profile Disconfirmation	7.32	5.62
	W16: Ownership, Sampling, and Comparative Experiences	6.60	5.54
	W17: Ambivalence and Qualified Evaluation	4.53	4.69
	Nb. of Sentences	23,631	20,559
Unisex	U1: Wearability and Experiential Appeal	5.31	4.65
	U2: Balanced Olfactory Recombination (Warm and Resinous Accords)	8.80	8.10
	U3: Ownership, Sampling, and Value-Based Evaluation	4.00	3.60
	U4: Balanced Olfactory Recombination (Woody and Aromatic Accords)	8.40	9.50
	U5: Perceived Artistry and Distinctive Appeal	12.75	16.70
	U6: Qualified Evaluation and Mild Disappointment	8.15	6.22
	U7: Perceived Artistry and Compositional Excellence	8.88	8.12
	U8: Gender Perception (Directional Lean)	6.24	7.03
	U9: Balanced Olfactory Recombination (Oud Integration)	3.49	3.47
	U10: Negative Expectation Disconfirmation (Conceptual Ambiguity)	8.35	8.20
	U11: Balanced Olfactory Recombination (Temporal Evolution)	4.71	4.54
	U12: Balanced Olfactory Recombination (Layered Integration)	5.65	5.92
	U13: Performance Intensity and Wearability	2.61	2.88
	U14: Comparative Evaluation (Scent Similarity)	5.89	5.77
	U15: Ownership, Sampling, and Comparative Experiences	6.78	5.29
	Nb. of Sentences	11,687	13,988

Note. Entries report the percentage of sentences assigned to each theme within each segment. Exploratory percentages are based on the top 10% most discriminative sentences, whereas confirmatory percentages are based on nearest-neighbor assignment of sentences in the confirmatory sample. Percentages sum to 100% within each segment.

“mystical,” or “very lovely” (W1, W7). Together, these themes suggest that atypical fragrances evoke not only distinctive olfactory impressions but also heightened sensory, emotional, and aesthetic responses.

Finally, men’s and women’s themes reveal blurred category boundaries. Men’s atypical fragrances are frequently described as “quite unisex” or “slightly feminine” (M11). In women’s fragrances, category blurring arises both through cross-gender recombination and explicit references to unisex appeal. In particular, spicy, woody, and resinous accords traditionally associated with masculine fragrances are combined with feminine elements (e.g., “the scent is spicy, sweet, and woody”; W6), while other themes explicitly characterize these fragrances as unisex (e.g., “definitely a unisex scent”; W9). These themes are consistent with movement toward a shared olfactory space that partially transcends traditional gender categories.

In contrast, the themes associated with *typical* unisex fragrances emphasize balanced olfactory recombination, compositional integration, and refinement rather than cross-gender recombination and boundary expansion. Several themes focus on the harmonious integration of woods, spices, incense, amber, and oud, often described in terms of temporal evolution and interactions among accords (e.g., “it’s an interesting mix of sweetness, spice, and wood,” “as the fragrance dries down, it becomes a little darker and woodsier,” and “this gives me a blast of lemon at first, then it becomes complex with fresh spicy, aromatic notes”; U2, U4, U9, U11, U12). Rather than emphasizing distinctive materials or departures from conventional category boundaries, these themes highlight compositional coherence and the balanced evolution of fragrances over time.

Additional themes emphasize refinement, compositional quality, and experiential appeal. Reviewers describe these fragrances as “a work of art,” “an astounding composition,” and “a fragrance with personality” (U5, U7), while other themes highlight wearability, versatility, performance, and broad consumer appeal (U1, U13). Themes involving slight masculine or feminine leanings are also present (U8), but these fragrances are typically described as balanced rather than explicitly gendered. These themes suggest that typical unisex fragrances derive their appeal from compositional coherence, refinement, and balanced integration while allowing for subtle directional positioning within a shared olfactory space.

Table 4 summarizes the relative frequency of the discovered themes among the top 10% most discriminative sentences retained in the exploratory sample. Theme frequencies vary substantially across segments, with men’s fragrances dominated by positive evaluation and tobacco- and leather-based recombination themes, women’s fragrances by hedonic richness, novelty, and gourmand recombination, and unisex fragrances by artistry, balanced olfactory recombination, and compositional excellence.

In summary, the exploratory analysis reveals distinct semantic patterns across fragrance segments. Atypical men’s and women’s fragrances are characterized by material-centered and cross-gender recombination, the intensification or reinterpretation of familiar accords, heightened sensory and emotional responses, and explicit or implicit blurring of traditional gender boundaries. In contrast, typical unisex fragrances emphasize balanced olfactory recombination, compositional integration, refinement, and coherent temporal evolution. These patterns suggest that consumers interpret

atypicality differently across segments, rewarding recombination and boundary expansion in the gendered segments while favoring balance and compositional coherence in the unisex segment. We next examine whether these themes reproduce in an independent sample and remain systematically associated with atypicality.

4.2 Confirmatory Phase

The confirmatory analysis consists of three stages. First, each sentence in the confirmatory sample is assigned to the theme of its nearest neighbor in the exploratory sample using cosine distance between sentence embeddings. Theme assignments are then aggregated to the review level to construct measures of theme prevalence. Second, we compare theme prevalence across atypicality quartiles. Third, we estimate segment-specific regression models relating membership in the focal fragrance group (Q_4 for men’s and women’s fragrances; Q_1 for unisex fragrances) to review-level theme prevalence while controlling for the full set of fragrance characteristics used in the structured-data analysis. Together, these analyses assess both the out-of-sample reproducibility of the discovered themes and their relationship to directional atypicality.

Out-of-Sample Theme Assignment: Each sentence in the confirmatory sample is assigned to the theme of its nearest neighbor in the exploratory sample using cosine distance between sentence embeddings. Sentences whose nearest neighbor belongs to the residual category (i.e., the lower 90% of sentences ranked by discriminative probability) are assigned to “Other Themes.” This results in 14,025, 20,559, and 13,988 sentences being assigned to the discovered themes for the men’s, women’s, and unisex segments, respectively, corresponding to 10.84%, 8.85%, and 11.87% of the confirmatory sentences, closely matching the proportion retained during the exploratory phase. Sentence-level theme assignments are then aggregated to the review level to construct review-level measures of theme prevalence used in the subsequent descriptive and regression analyses.

Table 4 compares the distribution of sentences assigned to the discovered themes in the exploratory and confirmatory samples, excluding the residual “Other Themes” category. Additional details of the assignment procedure are provided in Web Appendix K. The two distributions are quite similar across all three segments, with the most prevalent themes in the exploratory sample remaining the most prevalent in the confirmatory sample. Consistent with this visual similarity, Spearman correlations between the two distributions range from 0.939 to 0.969 across the three segments (all $p < 0.001$), indicating a high degree of out-of-sample reproducibility. These results indicate that the discovered themes are highly reproducible in previously unseen reviews.

Table 5 provides qualitative evidence of the out-of-sample reproducibility of the discovered themes by comparing representative sentences from the exploratory and validation samples. Although the sentences are not identical, they exhibit strong semantic similarity that extends well beyond shared keywords. For example, in men’s fragrances, the exploratory sentence “There’s some oud in the mix” corresponds closely to the validation sentence “Nice scent. The spices are wrapped around the oud,” with both expressing material-centric recombination centered on oud (M2). Similarly, in women’s fragrances, the exploratory sentence “There is something slightly smoky, airy about it” aligns with the validation sentence “There is something smokey that adds sophistication,” illustrating evocative depth and atmospheric complexity (W1). Finally, among

unisex fragrances, the exploratory sentence “Can be worn in most occasions” corresponds to the validation sentence “This could be worn in autumn, winter or spring,” with both emphasizing wearability and broad experiential appeal (U1). More examples and complete theme descriptions are provided in Web Appendix K.

Table 5: Illustrative Evidence of Out-of-Sample Theme Reproducibility

Theme	Exploratory Sample	Confirmatory Sample
Men		
M2: Material-Centric Recombination (Oud)	- There’s some oud in the mix. - I also get a hint of oud. - There is quite some oud lingering through.	- There is something powdery Vanilla/ Coffee/Patchouli/Cashmere about this Oud. - Nice scent The spices are wrapped around the oud. - this one is a dry/spicy oud Its a nice fragrance that oud, woody notes, cinnamon, vetiver and pink pepper are present.
M7: Cross-Gender Recombination (Sweet and Gourmand Accords)	- There is a sweetness in there as well. - It’s slightly sweet and musky. - There’s a subtle sweetness in there, too.	- And there is a bit of sweetness to it. - The sweetness is almost gooey. - There is hints of sweetness in here as well.
M12: Within-Category Recombination (Tobacco and Smoky Accords)	- Giving the tobacco a sweet and smoky vibe. - It’s aromatic, almost delicious. - The mix of smoke, sweet and leather is very attractive.	- There’s a subtle whiff of tobacco. - I also smell incense in this, maybe it is the rum and woody notes together. - Either this, or there is a dry tobacco accord in the scent.
Women		
W1: Evocative Depth and Atmospheric Complexity	- There is something slightly smoky, airy about it. - There is also a hint of leather in there. - It has a slight smokiness in it.	- Hint of vanilla/patchouli/leather at the end. - There is a slightly waxy greenness. - There is something smokey that adds sophistication.
W6: Cross-Gender Recombination (Spicy and Resinous Accords)	- It’s herbal and aromatic. - I like how there’s a tinge of incense in there as well. - There is some spice, some jasmine, some sweetness.	- Almost gourmand by scent. - Incense-y, herbal, but also floral. - There’s an herbal feel to it.
W9: Gender Boundary Blurring (Unisex Perception)	- This fragrance is UNISEX. - It is very unisex. - Definitely a unisex type of scent.	- It’s a rather unisex scent. - It is def a unisex fragrance. - This is also a unisex fragrance.
Unisex		
U1: Wearability and Experiential Appeal	- Makes for a more intimate wearing. - Can be worn in most occasions. - It compliments it actually.	- This could be worn in autumn, winter or spring. - It’s warm but has some different facets. - It feels good to have on.
U2: Balanced Olfactory Recombination (Warm and Resinous Accords)	- i like the hint of sandalwood. - A touch of something herbal and some woods thrown in as well. - It is a beautiful fruity scent with some herbs.	- There are woody and warm, slightly powdered (amber) sweet touches, too. - There’s a touch of powdery sandalwood. - The base is a wood and there is some spice to it as well.
U8: Gender Perception (Directional Lean)	- Unisex, but leans a bit masculine. - Unisex, but leans masculine. - Leaning a little masculine but seems unisex.	- Unisex, leans masculine in my opinion. - It is unisex, maybe leans a bit feminine. - This leans masculine, but is unisex.

Note. The table reports representative sentences from the exploratory and confirmatory samples for selected themes. Validation sentences are assigned the theme label of their nearest neighbor in the exploratory sample based on cosine distance between sentence embeddings. The examples illustrate that the discovered themes remain semantically consistent in previously unseen reviews.

Descriptive Analysis: Using the review-level measures of theme prevalence constructed in the previous step, we compare theme prevalence between highly atypical (Q_4) and highly typical (Q_1) fragrances within each segment. If the discovered themes generalize to previously unseen reviews, they should be more prevalent in the focal fragrance group (Q_4 for men’s and women’s fragrances; Q_1 for unisex fragrances). Table 6 reports the resulting descriptive comparisons together with the corresponding controlled regression analyses. For readability, we report only the core themes discussed in the exploratory analysis; complete results for all discovered themes are provided in Web Appendix L.

Table 6 reports the average review-level prevalence of each theme across the highest (Q_4) and lowest (Q_1) atypicality quartiles, together with difference-in-means tests. Positive differences indicate themes associated with highly atypical men’s and women’s fragrances, whereas negative differences indicate themes associated with highly typical unisex fragrances.

The descriptive patterns closely mirror the exploratory findings. For men’s fragrances, highly atypical fragrances (Q_4) exhibit significantly greater prevalence of themes involving material-centered recombination through oud, leather, and tobacco accords, together with cross-gender recombination through sweet and gourmand elements. Similarly, highly atypical women’s fragrances exhibit significantly greater prevalence of themes involving atmospheric complexity, novelty and distinctiveness, floral and gourmand intensification, cross-gender recombination, and gender-boundary blurring. These findings provide initial evidence that the semantic patterns identified in the exploratory analysis generalize to an independent sample.

In contrast, the unisex segment exhibits the opposite pattern. Highly typical fragrances (Q_1) exhibit significantly greater prevalence of themes emphasizing balanced olfactory recombination through warm and resinous, woody and aromatic, and oud-based accords, together with temporal evolution and layered integration. Themes highlighting artistry, compositional excellence, and broad wearability are also significantly more prevalent among highly typical fragrances. These findings suggest that, unlike the gendered segments, typical unisex fragrances are characterized by balance, integration, and refinement rather than by recombination and boundary expansion.

Overall, the descriptive patterns are consistent with the directional movements identified in the structured-data analysis. Reviews of highly atypical fragrances in the gendered segments emphasize recombination, intensification, and boundary expansion, whereas reviews of highly typical unisex fragrances emphasize balance, integration, and compositional coherence. These findings suggest that consumers interpret atypicality differently across segments, providing a behavioral explanation for the directional patterns observed in the structured-data analysis. We next examine whether these associations persist after accounting for fragrance characteristics and the coexistence of multiple themes within reviews.

Controlled Analysis: We estimate segment-specific linear probability models in which membership in the highest atypicality quartile (Q_4 , relative to Q_1) is regressed on review-level theme prevalence measures and the full set of fragrance-level controls used in the structured-data analysis. Standard errors are clustered at the brand level. For men’s and women’s fragrances, positive coefficients indicate themes associated with highly atypical fragrances, whereas for unisex fragrances,

Table 6: Review-Level Theme Prevalence Across Atypicality Quartiles and Controlled Analysis

Theme	Descriptive Analysis				Controlled Analysis		
	Q_4	Q_1	Δ	p_{diff}	β	SE	p_{reg}
Men							
M2: Material-Centric Recombination (Oud)	0.040	0.007	0.033	< 0.001	0.132	0.037	< 0.001
M3: Intense Positive Evaluation and Aesthetic Appeal	0.107	0.061	0.046	< 0.001	0.032	0.010	0.001
M4: Material-Centric Recombination (Leather)	0.074	0.012	0.062	< 0.001	0.107	0.015	< 0.001
M6: Hedonic Richness and Emotional Appeal	0.051	0.028	0.024	< 0.001	0.035	0.010	< 0.001
M7: Cross-Gender Recombination (Sweet and Gourmand Accords)	0.070	0.029	0.041	< 0.001	0.035	0.009	< 0.001
M8: Intensity and Sensory Presence	0.065	0.037	0.028	< 0.001	0.038	0.008	< 0.001
M11: Gender Boundary Blurring (Unisex Perception)	0.027	0.018	0.009	< 0.001	0.003	0.016	0.863
M12: Within-Category Recombination (Tobacco and Smoky Accords)	0.066	0.018	0.048	< 0.001	0.063	0.014	< 0.001
Women							
W1: Evocative Depth and Atmospheric Complexity	0.041	0.018	0.022	< 0.001	0.073	0.009	< 0.001
W2: Within-Category Recombination (Rose Intensification)	0.024	0.008	0.016	< 0.001	0.012	0.018	0.501
W4: Within-Category Recombination (Creamy Gourmand Accords)	0.055	0.022	0.033	< 0.001	0.058	0.012	< 0.001
W5: Within-Category Recombination (Coconut Gourmand Accords)	0.026	0.005	0.020	< 0.001	0.114	0.026	< 0.001
W6: Cross-Gender Recombination (Spicy and Resinous Accords)	0.045	0.020	0.026	< 0.001	0.089	0.014	< 0.001
W7: Novelty and Distinctiveness	0.073	0.034	0.039	< 0.001	0.053	0.008	< 0.001
W9: Gender Boundary Blurring (Unisex Perception)	0.012	0.005	0.007	< 0.001	0.083	0.020	< 0.001
W10: Within-Category Recombination (Floral Intensification)	0.033	0.007	0.026	< 0.001	0.029	0.022	0.197
W11: Perceived Artistry and Refined Appeal	0.036	0.019	0.016	< 0.001	0.054	0.009	< 0.001
W12: Intense Positive Affect and Enthusiastic Evaluation	0.041	0.023	0.017	< 0.001	0.039	0.009	< 0.001
W13: Within-Category Recombination (White Floral Intensification)	0.035	0.015	0.020	< 0.001	-0.021	0.008	0.009
W14: Evocative Sensory Appeal and Hedonic Richness	0.071	0.038	0.033	< 0.001	0.022	0.007	< 0.001
Unisex							
U1: Wearability and Experiential Appeal	0.024	0.034	-0.010	< 0.001	0.001	0.011	0.966
U2: Balanced Olfactory Recombination (Warm and Resinous Accords)	0.028	0.072	-0.043	< 0.001	-0.062	0.011	< 0.001
U4: Balanced Olfactory Recombination (Woody and Aromatic Accords)	0.042	0.076	-0.034	< 0.001	-0.043	0.012	< 0.001
U5: Perceived Artistry and Distinctive Appeal	0.081	0.127	-0.046	< 0.001	-0.015	0.006	0.012
U7: Perceived Artistry and Compositional Excellence	0.038	0.063	-0.024	< 0.001	0.005	0.009	0.546
U8: Gender Perception (Directional Lean)	0.034	0.054	-0.020	< 0.001	-0.015	0.009	0.086
U9: Balanced Olfactory Recombination (Oud Integration)	0.018	0.025	-0.007	0.003	-0.011	0.015	0.486
U11: Balanced Olfactory Recombination (Temporal Evolution)	0.021	0.035	-0.014	< 0.001	-0.032	0.013	0.010
U12: Balanced Olfactory Recombination (Layered Integration)	0.025	0.049	-0.024	< 0.001	-0.032	0.010	0.002
U13: Performance Intensity and Wearability	0.016	0.020	-0.003	0.079	0.007	0.014	0.620

Note. Q_1 and Q_4 denote the lowest and highest atypicality quartiles, respectively. Columns Q_4 and Q_1 report the average review-level prevalence of each theme. Δ reports the difference in mean prevalence ($Q_4 - Q_1$), and p_{diff} reports the corresponding two-sided Welch test. Columns β , SE, and p_{reg} report estimates from segment-specific linear probability models in which Q_4 membership is regressed on review-level theme prevalence and fragrance-level controls. For men's and women's fragrances, positive coefficients indicate themes associated with highly atypical fragrances. For unisex fragrances, negative coefficients indicate themes associated with highly typical fragrances.

negative coefficients indicate themes associated with highly typical fragrances (Q_1).

Correlations among the theme prevalence measures are uniformly small, ranging from 0.000 to 0.097 for men, -0.008 to 0.081 for women, and -0.002 to 0.085 for unisex fragrances (see Web Appendix L), indicating that the themes capture distinct aspects of review content rather than highly overlapping semantic patterns. Together with the fragrance-level controls, these low correlations allow the estimated coefficients to be interpreted as the associations between individual themes and atypicality, conditional on the remaining themes and observable fragrance characteristics.

For men’s fragrances, the strongest positive associations with atypicality are observed for material-centered recombination involving oud ($\beta = 0.132, p < 0.001$) and leather ($\beta = 0.107, p < 0.001$), followed by tobacco and smoky accords ($\beta = 0.063, p < 0.001$). Cross-gender recombination through sweet and gourmand elements also remains positively associated with atypicality ($\beta = 0.035, p < 0.001$). In contrast, explicit references to unisex positioning are no longer significantly associated with atypicality after controlling for the coexistence of other themes and fragrance characteristics ($\beta = 0.003, p = 0.863$). These results indicate that atypical men’s fragrances are characterized primarily by olfactory recombination through the reinterpretation of iconic masculine accords and the incorporation of cross-gender olfactory elements, rather than by explicit perceptions of gender-boundary blurring.

Women’s fragrances exhibit a different pattern. The strongest positive association with atypicality is observed for coconut gourmand intensification ($\beta = 0.114, p < 0.001$), followed by cross-gender recombination through spicy and resinous accords ($\beta = 0.089, p < 0.001$) and explicit gender-boundary blurring ($\beta = 0.083, p < 0.001$). Atmospheric complexity also remains strongly associated with atypicality ($\beta = 0.073, p < 0.001$). These results indicate that atypical women’s fragrances are characterized by both the intensification of traditionally feminine accords and the incorporation of cross-gender olfactory elements, resulting in more distinctive compositions that blur conventional gender boundaries.

The unisex segment exhibits the opposite pattern. For unisex fragrances, negative coefficients indicate themes associated with highly *typical* fragrances (Q_1). The strongest negative associations with atypicality are observed for balanced olfactory recombination through warm and resinous ($\beta = -0.062, p < 0.001$) and woody and aromatic accords ($\beta = -0.043, p < 0.001$). Themes emphasizing temporal evolution ($\beta = -0.032, p = 0.010$) and layered integration ($\beta = -0.032, p = 0.002$) are also negatively associated with atypicality. These results indicate that typical unisex fragrances are characterized by balanced olfactory recombination, coherent temporal evolution, and refined compositional integration, rather than by the cross-category recombination and boundary expansion that characterize the gendered segments.

Overall, the textual analyses provide a consumer-level interpretation of the directional patterns identified in the structured-data analysis. In the gendered segments, atypicality is associated with material-centered and cross-gender recombination, the intensification of familiar accords, atmospheric complexity, and, particularly for women’s fragrances, explicit gender-boundary blurring. In contrast, the unisex segment is characterized by balanced olfactory recombination, coherent temporal evolution, and refined compositional integration. These semantic patterns closely mirror

the directional movements observed in category space, indicating that consumers reward atypicality when it moves fragrances toward interpretable cross-segment positions but penalize it when it disrupts the coherence of a balanced unisex prototype.

5 Discussion

We examine how consumers evaluate differentiation in markets characterized by multiple local prototypes. Using large-scale data on fragrances and consumer reviews, we show that the value of atypicality depends not only on the magnitude of deviation from a category prototype but also on its direction relative to those prototypes. Specifically, atypicality is associated with higher evaluations for men’s and women’s fragrances but lower evaluations for unisex fragrances. This asymmetry reflects *directional atypicality*: consumer evaluations are more favorable when deviations position products within valued and interpretable cross-segment regions, and less favorable when deviations distance products from coherent segment prototypes.

Our findings further demonstrate that these patterns emerge only when atypicality is measured relative to segment-specific prototypes. Consider the men’s fragrance discussed above. Adding sweet edible notes may reposition it toward a neighboring unisex prototype and increase its appeal. By contrast, a unisex fragrance that becomes increasingly distinctive by emphasizing unusual niche accords moves away from the balanced unisex prototype, reducing rather than enhancing its appeal. These contrasting examples illustrate why there is no single global optimum in segmented markets. Rather, the value of differentiation depends on how products move within category space. When fragrances are pooled, atypicality is defined relative to a common benchmark that conflates distinct segment structures, producing a weak aggregate relationship that masks the underlying evaluative dynamics. Aggregate measures of atypicality therefore provide only a partial account of consumer evaluations.

We characterize these patterns by mapping fragrances into a common category space and examining how products move as they become more atypical. This representation shows that differentiation follows systematic directional trajectories rather than diffuse departures from category norms. Directional atypicality therefore characterizes products by their trajectories through category space rather than their distance from a single prototype. As atypicality increases, men’s and women’s fragrances converge toward a shared cross-segment region, whereas unisex fragrances remain anchored near a balanced central prototype. Once these directional movements are incorporated into the analysis, the direct association between atypicality and consumer evaluations largely disappears. This finding suggests that consumers do not reward differentiation per se, but rather differentiation that moves products toward valued positions within category space. The moderate correlation between atypicality and directional atypicality further indicates that directional measures capture information beyond scalar distance from a category prototype.

The textual analyses reveal how consumers interpret these directional movements. Returning to the example introduced at the beginning of the paper, consumers do not simply perceive a men’s fragrance with sweet edible notes as more unusual. Rather, they interpret it as a meaningful recombination of recognizable masculine and feminine olfactory elements. More broadly, in the

gendered segments, consumers reward specific forms of recombination rather than novelty or differentiation alone. Men's fragrances are characterized by material-centered recombination involving iconic masculine accords such as oud, leather, and tobacco together with cross-gender olfactory elements, whereas women's fragrances combine the intensification of traditionally feminine scents with cross-gender recombination that produces more distinctive and atmospheric compositions. By contrast, consumers evaluate unisex fragrances relative to a balanced central prototype, rewarding balanced olfactory recombination, coherent temporal evolution, refined compositional integration, and subtle directional positioning rather than stronger departures from the category center. These findings suggest that consumers assign different meanings to similar levels of atypicality depending on where products move within category space. Directional movements create value not simply by increasing novelty, but by introducing novelty in ways that remain interpretable, allowing consumers to recognize meaningful departures from category expectations rather than arbitrary deviations.

Our findings refine research on categorization, schema congruity, and creativity by demonstrating that category structure fundamentally shapes the value of differentiation. Prior research typically conceptualizes atypicality as a scalar construct defined relative to a single category benchmark. Our results suggest instead that differentiation is inherently relational: products are evaluated relative to multiple local prototypes whose positions define the surrounding category structure. Consequently, the meaning of a given level of atypicality depends not only on its magnitude but also on the neighboring prototype toward which it moves. The men's fragrance introduced at the beginning of the paper illustrates this broader principle. Whether consumers interpret the addition of sweet edible notes as creative or inappropriate depends less on its distance from the traditional masculine prototype than on the neighboring prototype toward which it moves. In doing so, our framework extends research on positioning relative to category knowledge (Sujaan and Bettman, 1989) by conceptualizing differentiation not as increasing distance from a single category reference point, but as movement through a structured system of related categories in which neighboring prototypes jointly shape consumer evaluations.

This relational perspective has three broader theoretical implications. First, it advances prototype-based accounts of categorization (Loken, 2006; Loken and Ward, 1990; Ratneshwar and Shocker, 1991; Rosch, 1978) by showing that category judgments are shaped not only by proximity to a prototype but also by the configuration of neighboring prototypes within the broader category structure. It also complements research examining multiple-category inference and activation under categorization uncertainty (Gregan-Paxton, Hoeffler, and Zhao, 2005; Lajos et al., 2009) by shifting attention from how consumers simultaneously draw on multiple categories to how products move through a structured system of local prototypes. Second, it complements research on creative recombination (Burroughs, Moreau, and Mick, 2018; Toubia and Netzer, 2017; Uzzi et al., 2013) by showing that consumers do not reward novelty or recombination in itself. Rather, successful differentiation arises when recombination repositions products toward neighboring prototypes while preserving their interpretability. Finally, it extends research on category spanning (Hsu, 2006; Pontikes, 2012; Zuckerman, 1999) by showing that the value of spanning depends not only on whether products cross category boundaries, but also on which neighboring category they move toward and how that movement reshapes their relationship to the surrounding category structure.

Our findings also provide a richer understanding of how consumers evaluate fragrances. Rather

than rewarding novelty alone, consumers appear to value fragrances that reinterpret familiar olfactory profiles in ways that remain meaningful and recognizable. A men's fragrance, for example, becomes more appealing not simply because it incorporates sweeter edible notes, but because those additions create a coherent movement toward a neighboring unisex prototype while preserving its masculine identity. Similarly, successful women's fragrances combine traditionally feminine scents with carefully selected masculine cues, whereas successful unisex fragrances derive their appeal from maintaining balance and compositional coherence rather than maximizing distinctiveness. Together, these patterns suggest that fragrance evaluation reflects consumers' ability to recognize and appreciate meaningful movements within a structured olfactory category system rather than departures from category norms alone. More broadly, they suggest that hybrid categories such as unisex fragrances need not remain ambiguous mixtures of existing categories. Instead, they may develop coherent local prototypes that consumers evaluate according to their own standards of balance, integration, and interpretability.

Although our empirical setting is fragrances, the proposed framework applies more broadly to markets in which products can be represented within a common feature space and evaluated relative to multiple local prototypes. Many such markets contain hybrid categories that occupy stable positions between established segments and serve as coherent reference points for consumer evaluation. In music, for example, jazz fusion and classical crossover have evolved into recognized genres with their own conventions rather than representing arbitrary combinations of musical styles. Similarly, gender-neutral fashion occupies a distinct position between traditional menswear and womenswear, while plant-based meat alternatives have emerged as a recognizable category between conventional meat and vegetarian products. Media markets exhibit similar dynamics, with genres such as dramedies and docudramas combining familiar elements while establishing coherent identities of their own. Across these settings, successful differentiation depends not simply on combining familiar and novel elements, but on how products are repositioned relative to neighboring category prototypes while remaining interpretable to consumers.

5.1 Managerial Implications

Our findings suggest that managers should evaluate differentiation relative to the competitive structure of the market rather than broad category norms. In segmented markets, the same degree of differentiation can generate very different consumer responses depending on the direction of movement within category space. For example, adding sweet edible notes to a men's fragrance may create value not because it maximizes novelty, but because it moves the product toward a neighboring unisex position while remaining readily interpretable to consumers. Consequently, managers should evaluate new product concepts not only in terms of how different they are from existing offerings, but also in terms of how they reposition products relative to neighboring segments. Successful differentiation depends less on maximizing novelty than on identifying directions of movement that consumers perceive as meaningful and valuable.

These insights also have implications for creativity, innovation, and ideation. Many innovation processes implicitly assume that greater novelty leads to greater value. Our results suggest instead that the value of novelty depends on how new ideas recombine existing category elements while remaining interpretable to consumers. For fragrance managers, this may involve introducing

gourmand notes into a men’s fragrance, incorporating woody or resinous accords into a women’s fragrance, or strengthening balance and integration in a unisex fragrance. In each case, the objective is not simply to create a more unusual scent, but to reposition the fragrance toward a valued neighboring prototype while preserving its coherence. Managers engaged in ideation may therefore benefit from treating innovation as a search for meaningful recombinations rather than maximum novelty.

The findings also suggest that successful differentiation requires different innovation strategies across market segments rather than a single innovation philosophy. In established segments such as men’s and women’s fragrances, firms can create value by selectively incorporating cues from neighboring segments while maintaining a recognizable category identity. In contrast, hybrid segments such as unisex fragrances derive value from reinforcing balance, integration, and compositional coherence rather than pursuing increasingly distinctive offerings. Managers overseeing brands that compete across multiple segments should therefore tailor differentiation strategies to the local structure of each segment instead of applying the same degree or type of novelty across their product lines.

5.2 Limitations and Future Research

Our findings should be interpreted in light of several limitations that also suggest directions for future research. First, despite extensive controls, our analysis is observational and therefore does not establish causality. Future research could complement our findings with controlled experiments that manipulate the direction of differentiation while holding constant its magnitude, for example, by varying whether products move toward neighboring category prototypes, more extreme versions of the focal prototype, or balanced hybrid positions. Such studies would provide a useful extension by directly examining the causal mechanisms underlying directional atypicality.

Second, our analysis relies on user-generated reviews from a single platform, which may reflect the preferences of more engaged consumers and limit external validity. Although we mitigate observable selection differences through propensity-based balancing, *Fragrantica* reviews are available for only 67% of the fragrances in our sample, and unobserved selection effects may remain. In addition, the semantic themes are discovered and validated using embedding-based methods applied to this review corpus. Although the exploratory and confirmatory analyses demonstrate that these themes are highly reproducible, their interpretation may depend on the linguistic conventions and reviewing practices of the platform. While our empirical implementation represents fragrances using industry-standard accords and consumer reviews, the proposed framework does not depend on this particular representation. Future research could examine its robustness across alternative feature representations, review platforms, languages, and consumer communities.

Third, our empirical setting focuses on fragrances, a market characterized by both well-defined and hybrid segments. Although this setting provides a natural context for studying directional atypicality, future research should examine the conditions under which similar evaluative dynamics emerge in other markets. In particular, characteristics such as the number and overlap of local prototypes, the strength of category boundaries, and the pace of innovation may influence the extent to which consumers reward directional rather than scalar forms of differentiation.

Finally, our analysis is static and treats category prototypes as fixed. Yet products do not merely move within category space; they may also reshape the prototypes that organize it. An important extension is to examine how local prototypes emerge, evolve, and are contested over time, and how these processes influence which forms of recombination and boundary crossing consumers perceive as legitimate or appealing. Future research could further examine heterogeneity in responses to directional atypicality, exploring whether experts, innovators, and mainstream consumers differ in the directional movements they reward (Hamilton, Puntoni, and Tavassoli, 2010). More broadly, our findings suggest that category space itself is dynamic rather than fixed, opening a research agenda on how local prototypes evolve and how consumers respond to directional positioning as category structures change over time.

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Web Appendix

These materials have been supplied by the authors to aid in the understanding of their paper. The AMA is sharing these materials at the request of the authors.

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A Web Appendix A: Data Construction and Descriptive Statistics

We provide additional details on the data sources, preprocessing procedures, feature construction, and descriptive statistics used in the analysis.

A.1 Data Sources

Our analysis combines two complementary datasets scraped from *Fragrantica*, an online platform where users rate, review, and discuss fragrances. The first and primary dataset⁴ contains fragrance-level information, including ratings, olfactory accords, perfumers, brands, target segments, and release years. The second dataset⁵ contains user-written reviews. The datasets were scraped independently in 2023 and 2024 and matched at the fragrance level using the website url of the fragrance, yielding a unified dataset linking fragrance characteristics, consumer ratings, and textual reviews.

After preprocessing, the dataset contains 61,868 unique fragrances with non-missing ratings. Ratings are measured on a five-point scale with an average of 3.98. Release years span multiple decades, with the largest share of fragrances originating from the 2010s and 2020s. Brand and perfumer representation is concentrated. *Avon* (8% of fragrances), *The Dua Brand* (7%), and *Zara* (6%) are the most prevalent brands in the dataset. Among perfumers, *Alberto Morillas* is the most prolific, accounting for approximately 2% of all fragrances.

A.2 Data Cleaning

We removed fragrances with missing ratings or accord information and eliminated duplicate fragrance while retaining the duplicate with the largest number of ratings. Product name and brand identifiers were standardized, and release years were extracted when available. Missing perfumer identities and release years were retained as separate covariates.

A.3 Feature Construction

Each fragrance is represented by a binary vector indicating the presence or absence of the 90 olfactory accords. Brand and perfumer (i.e., nez) are represented by indicator variables, with entities appearing in fewer than 20 fragrances pooled into an “Other” category. Temporal controls are constructed as decade indicators based on release year, including a separate category for missing release year.

⁴https://www.kaggle.com/datasets/olgagmiufana1/fragrantica-com-fragrance-dataset/data?select=fra_perfumes.csv

⁵https://github.com/rdemarqui/perfume_recommender/tree/main

A.4 Rating Distributions by Segment

Figure A.1 presents the distribution of consumer ratings across fragrance gender segments. The distributions are broadly similar across segments and are approximately bell-shaped, with ratings concentrated around four.

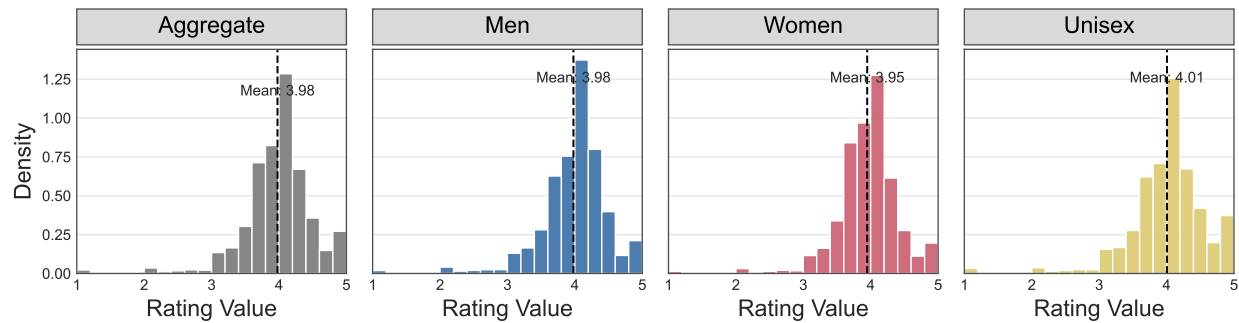


Figure A.1: Distribution of Consumer Ratings by Fragrance Segment

A.5 Distribution of Accords

Table A.1 reports the distribution of olfactory accords across target segments. Men's fragrances are characterized primarily by woody, aromatic, citrus, fresh spicy, and lavender accords, whereas women's fragrances emphasize floral, white floral, fruity, powdery, and sweet accords. Unisex fragrances combine characteristics of both segments, with relatively high frequencies of woody, amber, warm spicy, oud, smoky, and balsamic accords.

Table A.1: Distribution of Fragrance Accords by Gender (%)

# Accord	Agg.	Men	Women	Unisex	# Accord	Agg.	Men	Women	Unisex
1 Woody	65.666	85.027	56.604	66.522	46 Metallic	1.767	1.261	0.616	3.123
2 Citrus	48.852	66.800	49.329	40.821	47 Anis	1.400	2.587	1.084	1.214
3 Powdery	51.529	43.658	59.141	47.275	48 Coffee	1.466	1.650	0.844	2.007
4 Aromatic	42.909	78.212	25.020	45.830	49 Cacao	1.532	1.150	1.088	2.136
5 Sweet	47.996	28.806	60.335	43.808	50 Conifer	1.119	1.585	0.373	1.663
6 Warm Spicy	39.521	57.528	25.526	45.854	51 Chocolate	1.319	0.501	1.241	1.741
7 Fresh Spicy	35.846	67.282	20.127	38.237	52 Rum	0.923	1.122	0.338	1.421
8 Amber	35.516	41.813	25.746	42.578	53 Cherry	1.149	0.121	1.629	1.105
9 Floral	39.676	14.491	59.490	30.581	54 Soapy	0.786	0.269	0.958	0.832
10 Musky	34.317	29.714	37.043	33.544	55 Mineral	0.579	0.603	0.243	0.902
11 Fruity	35.060	19.887	51.315	25.287	56 Camphor	0.508	0.686	0.161	0.777
12 White Floral	32.923	14.148	48.850	24.994	57 Sour	0.584	0.074	1.068	0.316
13 Vanilla	28.234	19.785	33.167	26.888	58 Whiskey	0.398	0.566	0.055	0.668
14 Fresh	26.702	22.260	36.171	19.157	59 Bitter	0.322	0.334	0.204	0.433
15 Green	24.214	30.363	22.742	23.089	60 Savory	0.336	0.065	0.318	0.468
16 Earthy	17.280	26.627	11.304	19.286	61 Cannabis	0.281	0.250	0.035	0.539
17 Rose	21.357	8.029	29.724	18.650	62 Beeswax	0.317	0.065	0.204	0.535
18 Patchouli	13.821	17.161	12.380	13.848	63 Champagne	0.278	0.056	0.416	0.234
19 Balsamic	14.306	12.618	9.329	19.966	64 Terpenic	0.196	0.074	0.287	0.156
20 Animalic	11.450	10.337	8.976	14.379	65 Sand	0.181	0.093	0.149	0.250
21 Soft Spicy	9.814	11.302	10.621	8.386	66 Wine	0.181	0.083	0.079	0.324
22 Lavender	7.101	22.566	1.979	5.680	67 Alcohol	0.134	0.102	0.094	0.187
23 Leather	8.345	13.861	2.470	11.865	68 Vodka	0.100	0.176	0.086	0.082
24 Herbal	7.029	15.094	2.698	7.941	69 Oily	0.123	0.019	0.035	0.254
25 Aquatic	8.028	9.651	9.659	5.723	70 Coca-Cola	0.058	0.028	0.035	0.094
26 Violet	7.046	4.997	9.451	5.517	71 Clay	0.036	0.009	0.008	0.074
27 Mossy	5.584	10.300	3.522	5.649	72 Asphalt	0.023	0.037	0.000	0.039
28 Iris	6.899	4.376	7.979	6.887	73 Paper	0.029	0.000	0.008	0.062
29 Oud	6.334	4.654	1.382	11.966	74 Vinyl	0.023	0.019	0.020	0.027
30 Smoky	5.660	6.601	1.783	9.120	75 Plastic	0.016	0.000	0.000	0.039
31 Yellow Floral	6.414	1.048	9.270	5.833	76 Gasoline	0.013	0.009	0.000	0.027
32 Ozonic	4.998	6.694	5.316	3.967	77 Rubber	0.010	0.009	0.000	0.020
33 Tropical	5.602	2.485	8.689	3.846	78 Industrial Glue	0.006	0.000	0.000	0.016
34 Cinnamon	3.663	6.583	1.916	4.170	79 Hot Iron	0.006	0.000	0.000	0.016
35 Lactonic	4.907	0.640	7.064	4.560	80 Bacon	0.005	0.000	0.004	0.008
36 Marine	3.113	6.267	1.755	3.135	81 Sake	0.005	0.000	0.004	0.008
37 Tobacco	3.166	4.673	0.569	5.114	82 Tennis Ball	0.005	0.000	0.000	0.012
38 Tuberose	3.512	0.380	5.403	2.952	83 Pear	0.005	0.000	0.000	0.012
39 Honey	2.798	1.140	2.686	3.607	84 Foresty	0.002	0.009	0.000	0.000
40 Nutty	2.698	1.159	3.027	3.018	85 Milky	0.003	0.000	0.004	0.004
41 Caramel	2.798	0.714	3.663	2.815	86 Brown Scotch Tape	0.002	0.000	0.000	0.004
42 Almond	2.719	0.853	3.432	2.795	87 Meat	0.002	0.000	0.000	0.004
43 Coconut	2.397	0.621	3.047	2.499	88 Gourmand	0.002	0.000	0.000	0.004
44 Salty	1.818	2.995	1.072	2.065	89 Wet Plaster	0.002	0.000	0.000	0.004
45 Aldehydic	2.198	1.307	2.874	1.901	90 BBQ	0.002	0.000	0.000	0.004

Note. Values are percentages of fragrances containing each accord. Aggregate values are computed across all fragrances; gender-specific values are computed within each segment.

B Web Appendix B: Alternative Measures of Atypicality

To assess robustness, we re-estimate the main analyses using several alternative measures of olfactory atypicality. Each measure captures deviation from the segment-specific accord structure, but differs in whether atypicality is defined relative to all other fragrances in the segment, relative to the segment centroid, relative to local neighbors, or relative to a probabilistic model of accord occurrence.

Within each segment s , fragrance i is represented by an accord vector $\mathbf{x}_i \in \mathbb{R}^K$, where K denotes the number of possible accords and, for binary data, $x_{ik} \in \{0, 1\}$. Let N_s denote the number of fragrances in segment s , and let the segment centroid be

$$\boldsymbol{\mu}_s = \frac{1}{N_s} \sum_{j=1}^{N_s} \mathbf{x}_j.$$

Table B.1 summarizes the alternative atypicality measures used in the robustness analyses.

B.1 Results Across Alternative Atypicality Measures

Table B.2 reports the estimated effect of atypicality on consumer ratings across the alternative measures defined above. The results are highly consistent across specifications. For fragrances targeted to men and women, atypicality is positively associated with ratings across all measures. In contrast, the association is consistently negative for unisex fragrances. The similarity of the estimates across distance metrics indicates that the main findings are not driven by a particular operationalization of atypicality. The results demonstrate that the relationship between atypicality and consumer evaluations is robust to a broad range of geometric and probabilistic measures of atypicality.

Table B.1: Alternative Measures of Olfactory Atypicality

Metric family	Measure	Atypicality score	What it captures
<i>Panel A: Pairwise distance from other fragrances in the same segment</i>			
Angular distance	Cosine	$\frac{1}{N_s - 1} \sum_{\substack{j=1 \\ j \neq i}}^{N_s} \left(1 - \frac{\mathbf{x}_i^\top \mathbf{x}_j}{\ \mathbf{x}_i\ \ \mathbf{x}_j\ } \right)$	Deviation in the <i>orientation</i> of the accord profile, less sensitive to the number of accords.
Absolute distance	Euclidean	$\frac{1}{N_s - 1} \sum_{\substack{j=1 \\ j \neq i}}^{N_s} \ \mathbf{x}_i - \mathbf{x}_j\ _2$	Overall distance in accord space, sensitive to both accord composition and accord count.
Binary mismatch	Hamming	$\frac{1}{N_s - 1} \sum_{\substack{j=1 \\ j \neq i}}^{N_s} \left(\frac{1}{K} \sum_{k=1}^K \mathbf{1}\{x_{ik} \neq x_{jk}\} \right)$	Fraction of accord dimensions on which fragrance <i>i</i> differs from other fragrances.
<i>Panel B: Distance from the segment centroid</i>			
Angular distance to centroid	Cosine-centroid	$1 - \frac{\mathbf{x}_i^\top \boldsymbol{\mu}_s}{\ \mathbf{x}_i\ \ \boldsymbol{\mu}_s\ }$	Deviation from the average segment profile in terms of compositional orientation.
Absolute distance to centroid	Euclidean-centroid	$\ \mathbf{x}_i - \boldsymbol{\mu}_s\ _2$	Distance from the segment prototype in the full accord space.
<i>Panel C: Probabilistic and local-neighborhood measures</i>			
Probabilistic rarity	Negative log-likelihood	$-\log\left(\prod_{k=1}^K p_{sk}^{x_{ik}} (1 - p_{sk})^{1-x_{ik}}\right),$ with $p_{sk} = \frac{1}{N_s} \sum_{j=1}^{N_s} x_{jk}$.	How unlikely the accord profile is under segment-level accord frequencies.
Local distance	<i>m</i> -nearest-neighbors (KNN)	$\frac{1}{m} \sum_{j \in \mathcal{N}_m(i)} \left(1 - \frac{ A_i \cap A_j }{ A_i \cup A_j } \right)$	Deviation from the most similar fragrances in the same segment, rather than from the full segment.

Note. Fragrance *i* is represented by the binary accord vector $\mathbf{x}_i \in \mathbb{R}^K$, where $x_{ik} \in \{0, 1\}$. The segment centroid is $\boldsymbol{\mu}_s = N_s^{-1} \sum_{j=1}^{N_s} \mathbf{x}_j$. N_s is the number of fragrances in segment *s*, and $\mathcal{N}_m(i)$ is the set of the *m* nearest neighbors of fragrance *i* within segment *s*. For the nearest-neighbor measure, A_i denotes the set of accords contained in fragrance *i*. Higher values indicate greater deviation from the segment-specific accord structure. All measures are normalized to the $[0, 1]$ interval within each segment.

Table B.2: Linear Effects of Atypicality on Ratings Across Alternative Measures

Atypicality Measure	Men			Women			Unisex		
	Est.	SE	<i>p</i>	Est.	SE	<i>p</i>	Est.	SE	<i>p</i>
<i>Panel A: Pairwise distance from other fragrances in the same segment</i>									
Jaccard	0.124	0.056	0.026	0.116	0.029	< 0.001	-0.059	0.029	0.040
Cosine	0.128	0.061	0.035	0.121	0.030	< 0.001	-0.056	0.032	0.079
Euclidean	0.140	0.062	0.024	0.115	0.028	< 0.001	-0.074	0.035	0.036
Hamming	0.148	0.064	0.020	0.116	0.029	< 0.001	-0.078	0.037	0.034
<i>Panel B: Distance from the segment centroid</i>									
Cosine-Centroid	0.127	0.062	0.039	0.117	0.030	< 0.001	-0.061	0.033	0.060
Euclidean-Centroid	0.134	0.063	0.033	0.110	0.028	< 0.001	-0.071	0.036	0.051
<i>Panel C: Probabilistic and local-neighborhood measures</i>									
Neg. Log-Likelihood	0.165	0.074	0.027	0.154	0.036	< 0.001	-0.130	0.068	0.057
KNN (500)	0.113	0.063	0.071	0.138	0.033	< 0.001	-0.101	0.028	< 0.001
KNN (1,000)	0.118	0.061	0.052	0.147	0.034	< 0.001	-0.094	0.027	0.001
Controls									
# Ratings	Yes			Yes			Yes		
# Accords	Yes			Yes			Yes		
Perfumer	Yes			Yes			Yes		
Brand	Yes			Yes			Yes		
Decade	Yes			Yes			Yes		
Clustered SEs	Brand			Brand			Brand		
Observations	10,786			25,468			25,614		

Note. Standard errors clustered at the brand level are reported.

C Web Appendix C: Accord-Level Control Diagnostics

We provide additional justification for excluding accord-level controls from the baseline regressions, along with detailed multicollinearity diagnostics documenting the dependence of the atypicality measure on the underlying accord composition.

Atypicality is constructed directly from the underlying set of fragrance accords. Including the same accord indicators as controls would therefore condition on the variables that mechanically determine the regressor of interest, absorbing much of the variation used to identify the relationship between atypicality and consumer evaluations ([Angrist and Pischke, 2009](#)).

To quantify this dependence, we regress atypicality on the full set of 90 accord indicators separately for each segment. The resulting R^2 values are 0.98 for men's fragrances, 0.99 for women's fragrances, and 0.98 for unisex fragrances, indicating that atypicality is almost entirely explained by the underlying accord composition. Consistent with this mechanical relationship, including accord indicators in the rating regressions produces severe multicollinearity: the variance inflation factor (VIF) for atypicality equals 56.11 for men, 72.55 for women, and 55.04 for unisex fragrances, while the condition number of the design matrix ranges from 852 to 1,294, well above conventional diagnostic thresholds ([Belsley, Kuh, and Welsch, 2005](#); [Wooldridge, 2010](#)).

For this reason, the baseline specifications control for the number of accords rather than individual accord indicators, capturing overall compositional complexity without absorbing the variation that defines atypicality.

D Web Appendix D: Alternative Nonlinear Specifications

We assess the robustness of the nonlinear results to the discretization of atypicality. Complementing the quartile analysis reported in the main text, we re-estimate the models using terciles and quintiles.

Table D.1 reports the corresponding estimation results. Across both specifications, the results closely mirror those reported in the main text. For men’s and women’s fragrances, ratings increase monotonically with atypicality, with the largest and most statistically significant effects concentrated in the highest atypicality groups (T_3 and q_5). Intermediate groups generally exhibit smaller and often statistically insignificant effects.

For unisex fragrances, the pattern is reversed. Higher atypicality is associated with progressively lower ratings, with the strongest negative effects observed for the most atypical fragrances. In sum, the results show that consumer ratings are concentrated at the tails of the atypicality distribution: substantial departures from the segment prototype are associated with higher evaluations for men’s and women’s fragrances but lower evaluations for unisex fragrances.

Table D.1: Nonlinear Effects of Segment Atypicality on Ratings (Terciles and Quintiles)

	Men			Women			Unisex		
	Est.	SE	<i>p</i>	Est.	SE	<i>p</i>	Est.	SE	<i>p</i>
<i>Panel A: Terciles specification</i>									
T_1 (Most Typical)	—			—			—		
T_2	0.013	0.010	0.178	0.016	0.009	0.083	-0.009	0.006	0.153
T_3 (Most Atypical)	0.045	0.016	0.004	0.028	0.010	0.004	-0.027	0.010	0.008
Model Fit									
R^2		0.206			0.165			0.171	
Adj. R^2		0.136			0.130			0.135	
<i>Panel B: Quintiles specification</i>									
q_1 (Most Typical)	—			—			—		
q_2	0.014	0.014	0.327	0.011	0.008	0.194	-0.017	0.009	0.047
q_3	0.027	0.014	0.062	0.025	0.011	0.025	-0.001	0.008	0.903
q_4	0.028	0.019	0.141	0.022	0.009	0.011	-0.025	0.013	0.049
q_5 (Most Atypical)	0.066	0.028	0.017	0.049	0.014	< 0.001	-0.034	0.012	0.007
Model Fit									
R^2		0.206			0.166			0.171	
Adj. R^2		0.136			0.131			0.135	
Observations	10,786			25,468			25,614		
Controls									
# Ratings	Yes			Yes			Yes		
# Accords	Yes			Yes			Yes		
Perfumer	Yes			Yes			Yes		
Brand	Yes			Yes			Yes		
Decade	Yes			Yes			Yes		
Clustered SEs	Brand			Brand			Brand		

Note. Standard errors clustered at the brand level are reported. The omitted category corresponds to the most typical fragrances (T_1 and q_1).

E Web Appendix E: Weighted Aggregate Atypicality

We follow [Askin and Mauskopf \(2017\)](#) in the aggregate analysis of perfume atypicality. We first describe the procedure based on Jaccard similarity used to construct the aggregate atypicality measure in the main text. We then present a robustness analysis based on an alternative cosine-similarity specification.

E.1 Jaccard-Based Aggregate Atypicality

Each fragrance is represented as a binary vector indicating the presence or absence of olfactory accords. Pairwise similarities between fragrances are computed using the Jaccard similarity measure. Then, these similarities are weighted according to the cosine similarity between the aggregate accord profiles of the corresponding segments (men, women, and unisex). As a result, comparisons between fragrances belonging to more olfactorily similar segments receive greater weight.

The weight equals 1 when two fragrances belong to the same segment. Cross-segment weights are based on the cosine similarity between segment-level accord profiles: men–women = 0.77, men–unisex = 0.94, and women–unisex = 0.91.

For fragrance i , weighted atypicality is defined as the average weighted Jaccard distance to all other fragrances in the dataset:

$$\text{Atypicality}_i = \frac{1}{N-1} \sum_{j \neq i} w_{g_i g_j} d_J(i, j),$$

where N is the total number of fragrances in the dataset, $d_J(i, j)$ denotes the Jaccard distance between fragrances i and j , and $w_{g_i g_j}$ denotes the weight associated with the pair of market segments to which fragrances i and j belong. Finally, atypicality is normalized to range between 0 and 1, where higher values correspond to fragrances that are less similar to the overall market structure.

E.2 Cosine-Based Aggregate Atypicality

As a robustness check, we construct the aggregate atypicality measure using cosine similarity, closely following [Askin and Mauskopf \(2017\)](#). Table E.1 reports the corresponding estimation results. The results are qualitatively consistent with those reported in the main text. Across specifications, atypicality remains positively associated with consumer ratings at the aggregate level. In the linear specification, the coefficient on atypicality is positive and statistically significant. The quadratic specification also points to a positive relationship, although the quadratic term is not statistically significant. Similarly, the quartile specification shows that fragrances in higher atypicality quartiles receive significantly higher ratings relative to the most typical fragrances.

Table E.1: Aggregate Effects of Atypicality on Ratings

	Est.	SE	<i>p</i>
<i>Panel A: Linear specification</i>			
Atypicality	0.041	0.019	0.034
<i>Panel B: Quadratic specification</i>			
Atypicality	0.132	0.060	0.028
Atypicality ²	-0.118	0.076	0.123
<i>Panel C: Nonlinear specification</i>			
Q_1 (Most Typical)		—	
Q_2	0.012	0.006	0.034
Q_3	0.011	0.005	0.040
Q_4 (Most Atypical)	0.019	0.006	0.003
Observations		61,868	
Controls			
Segments		Yes	
# Ratings		Yes	
# Accords		Yes	
Perfumer		Yes	
Brand		Yes	
Decade		Yes	
Clustered SEs		Brand	

Note. Standard errors clustered at the brand level are reported. The quartile specification uses Q_1 (most typical fragrances) as the reference category.

F Web Appendix F: Predicting Fragrance Segments

We provide additional details on the multinomial logit (MNL) model used to predict fragrance segments. We first compare its predictive performance with that of more flexible machine-learning models and then report the estimated accord coefficients.

F.1 Model Performance

Table F.1 compares the predictive performance of the multinomial logit, a neural network, and a random forest using five-fold cross-validation. The multinomial logit performs comparably to the two machine-learning models across all evaluation metrics. Differences in accuracy, log loss, Brier score, and AUC are negligible, indicating that more flexible models provide little improvement in predictive performance.

Given its predictive performance, the multinomial logit provides a parsimonious and interpretable representation of segment structure while performing comparably to substantially more flexible classifiers. We therefore use its predicted probabilities throughout the analysis.

Table F.1: Model Performance Comparison for Gender Classification

	Accuracy	Log Loss	Brier Score	AUC
Multinomial Logit	0.650	0.765	0.465	0.812
Neural Net	0.653	0.757	0.461	0.816
Random Forest	0.653	0.766	0.465	0.816

F.2 Model Estimates

Tables F.2 and F.3 report the estimated multinomial logit coefficients, with unisex fragrances serving as the reference segment. The coefficients quantify how individual accords shift the probability that a fragrance is classified as women’s or men’s relative to unisex.

The estimates reveal a clear gender structure in accord composition. Floral, white floral, rose, and powdery accords are strongly associated with women’s fragrances, whereas woody, aromatic, warm spicy, and citrus accords are associated with men’s fragrances. Many accords exhibit opposite effects across the two gendered segments, reflecting their contrasting olfactory profiles. Conversely, accords such as oud, oily, savory, and cannabis decrease the probability of belonging to either gendered segment, increasing the likelihood of a unisex classification.

Table F.2: Multinomial Logit Model of Gender Segment on Olfactory Accords

Accord	Women			Accord	Men		
	Est.	SE	<i>p</i>		Est.	SE	<i>p</i>
Intercept	-0.422	0.035	< 0.001	Intercept	-2.229	0.054	< 0.001
Champagne	-0.225	0.190	0.236	Champagne	-1.173	0.464	0.012
Alcohol	-0.462	0.283	0.103	Alcohol	0.090	0.367	0.806
Aldehydic	0.348	0.072	< 0.001	Aldehydic	-0.203	0.113	0.072
Almond	-0.319	0.071	< 0.001	Almond	-0.293	0.133	0.028
Amber	-0.385	0.024	< 0.001	Amber	0.066	0.029	0.024
Animalic	-0.208	0.039	< 0.001	Animalic	-0.015	0.048	0.751
Anis	-0.258	0.099	0.009	Anis	0.715	0.101	< 0.001
Aquatic	0.242	0.049	< 0.001	Aquatic	0.409	0.061	< 0.001
Aromatic	-0.410	0.027	< 0.001	Aromatic	0.500	0.034	< 0.001
Balsamic	-0.102	0.036	0.004	Balsamic	-0.670	0.041	< 0.001
Beeswax	-0.432	0.191	0.024	Beeswax	-1.404	0.409	0.001
Bitter	-0.246	0.191	0.196	Bitter	-0.281	0.211	0.183
Cacao	-0.266	0.089	0.003	Cacao	-0.155	0.115	0.178
Camphor	-0.694	0.186	< 0.001	Camphor	-0.526	0.148	< 0.001
Cannabis	-1.388	0.365	< 0.001	Cannabis	-0.973	0.227	< 0.001
Caramel	0.168	0.059	0.005	Caramel	-0.689	0.129	< 0.001
Cherry	0.257	0.094	0.006	Cherry	-1.376	0.293	< 0.001
Chocolate	-0.017	0.091	0.853	Chocolate	-0.748	0.158	< 0.001
Cinnamon	-0.168	0.064	0.009	Cinnamon	0.352	0.059	< 0.001
Citrus	0.175	0.023	< 0.001	Citrus	0.688	0.029	< 0.001
Coconut	-0.201	0.068	0.003	Coconut	-0.436	0.146	0.003
Coffee	-0.499	0.095	< 0.001	Coffee	-0.034	0.100	0.734
Conifer	-0.559	0.124	< 0.001	Conifer	-0.422	0.100	< 0.001
Earthy	-0.276	0.037	< 0.001	Earthy	-0.023	0.037	0.536
Floral	0.759	0.023	< 0.001	Floral	-0.687	0.035	< 0.001
Fresh	0.210	0.027	< 0.001	Fresh	0.203	0.036	< 0.001
Fresh Spicy	-0.391	0.027	< 0.001	Fresh Spicy	0.383	0.031	< 0.001
Fruity	0.611	0.025	< 0.001	Fruity	-0.030	0.037	0.422
Green	-0.189	0.027	< 0.001	Green	0.046	0.032	0.150
Herbal	-0.340	0.052	< 0.001	Herbal	0.081	0.043	0.059
Honey	-0.509	0.063	< 0.001	Honey	-0.322	0.108	0.003
Iris	0.071	0.045	0.110	Iris	-0.543	0.064	< 0.001
Lactonic	0.095	0.049	0.052	Lactonic	-1.085	0.135	< 0.001
Lavender	-0.404	0.059	< 0.001	Lavender	0.942	0.041	< 0.001
Leather	-0.730	0.056	< 0.001	Leather	0.369	0.048	< 0.001

Table F.3: Multinomial Logit Model of Gender Segment on Olfactory Accords (Cont.)

Accord	Women			Accord	Men		
	Est.	SE	<i>p</i>		Est.	SE	<i>p</i>
Marine	-0.302	0.086	< 0.001	Marine	0.427	0.080	< 0.001
Metallic	-0.233	0.103	0.024	Metallic	-0.703	0.106	< 0.001
Mineral	-0.885	0.166	< 0.001	Mineral	-0.112	0.163	0.490
Mossy	0.026	0.056	0.635	Mossy	0.285	0.053	< 0.001
Musky	-0.250	0.024	< 0.001	Musky	-0.132	0.031	< 0.001
Nutty	-0.146	0.073	0.047	Nutty	-0.030	0.120	0.801
Oily	-1.937	0.373	< 0.001	Oily	-1.639	0.742	0.027
Oud	-1.814	0.063	< 0.001	Oud	-0.686	0.056	< 0.001
Ozonic	-0.104	0.058	0.072	Ozonic	0.267	0.068	< 0.001
Patchouli	0.336	0.036	< 0.001	Patchouli	0.145	0.039	< 0.001
Powdery	0.277	0.026	< 0.001	Powdery	0.136	0.032	< 0.001
Rose	0.382	0.027	< 0.001	Rose	-0.654	0.044	< 0.001
Rum	-1.016	0.134	< 0.001	Rum	-0.027	0.117	0.819
Salty	-0.226	0.107	0.034	Salty	-0.236	0.106	0.026
Sand	0.303	0.247	0.221	Sand	-0.999	0.365	0.006
Savory	-0.549	0.160	0.001	Savory	-1.103	0.428	0.010
Smoky	-0.424	0.062	< 0.001	Smoky	-0.137	0.055	0.013
Soapy	-0.520	0.105	< 0.001	Soapy	-0.449	0.213	0.035
Soft Spicy	0.168	0.038	< 0.001	Soft Spicy	0.149	0.046	0.001
Sour	0.628	0.140	< 0.001	Sour	-1.050	0.382	0.006
Sweet	0.196	0.026	< 0.001	Sweet	-0.148	0.035	< 0.001
Terpenic	0.282	0.220	0.199	Terpenic	-0.407	0.404	0.314
Tobacco	-1.410	0.096	< 0.001	Tobacco	0.106	0.065	0.102
Tropical	0.089	0.048	0.064	Tropical	0.277	0.083	0.001
Tuberose	0.200	0.056	< 0.001	Tuberose	-1.103	0.168	< 0.001
Vanilla	0.182	0.027	< 0.001	Vanilla	-0.081	0.036	0.025
Violet	0.171	0.044	< 0.001	Violet	0.261	0.064	< 0.001
Vodka	0.338	0.346	0.329	Vodka	0.620	0.365	0.090
Warm Spicy	-0.237	0.027	< 0.001	Warm Spicy	0.533	0.031	< 0.001
Whiskey	-1.355	0.293	< 0.001	Whiskey	-0.151	0.167	0.364
White Floral	0.739	0.024	< 0.001	White Floral	-0.535	0.036	< 0.001
Wine	-1.377	0.271	< 0.001	Wine	-0.915	0.378	0.016
Woody	-0.168	0.024	< 0.001	Woody	0.698	0.036	< 0.001
Yellow Floral	0.048	0.041	0.242	Yellow Floral	-1.278	0.104	< 0.001
Model Fit							
Observations	61,868						
Pseudo R^2	0.2636						
Log-Likelihood	-47,154						
LL-Null	-64,033						

Note. Standard errors are reported in the SE columns. The reference category is the unisex segment. Prior to estimation, accord indicators with near-zero variance were excluded to ensure numerical stability. Specifically, accords with prevalence below 0.1% were removed, reducing the total number of accords from 90 to 69.

G Web Appendix G: Linear Discriminant Analysis

We provide additional details on the linear discriminant analysis (LDA) used to visualize the fragrance space.

To focus on the accords most informative for segment separation, we retain only those that are statistically significant in at least one equation of the multinomial logit model at the 5% level. Using this reduced accord matrix, we estimate a two-dimensional LDA that maximizes between-segment relative to within-segment variation. The fitted LDA maps each fragrance to a two-dimensional space, from which we compute segment-specific centroids for the most typical (Q_1) and most atypical (Q_4) fragrances. Arrows connecting the two centroids illustrate the direction of atypicality shown in Figure 5 of the main text.

To facilitate interpretation, we compute the vector length of each accord loading,

$$\text{Norm}_k = (LD1_k^2 + LD2_k^2)^{1/2},$$

where larger values indicate stronger contributions to segment separation. Table G.1 reports the accords with the largest vector lengths. LD1 captures gender polarity, with floral, white floral, rose, and fruity loading positively and woody, aromatic, lavender, and fresh spicy loading negatively. LD2 captures compositional unconventionality, with accords such as oud, oily, wine, cannabis, and camphor contributing most strongly.

Table G.1: Top Accord Vectors in the LDA Space

Accord	LD1	LD2	Norm	Min. p-value
Oily	-0.732	2.180	2.299	< 0.001
Oud	-0.462	1.655	1.718	< 0.001
Wine	-0.459	1.629	1.692	< 0.001
Lavender	-0.863	-1.427	1.667	< 0.001
Cannabis	0.184	1.275	1.288	< 0.001
Metallic	0.193	0.991	1.010	< 0.001
Beeswax	0.085	1.002	1.006	< 0.001
Savory	-0.126	0.948	0.957	< 0.001
Camphor	-0.020	0.860	0.860	< 0.001
Sand	0.508	0.591	0.779	0.006
Rum	-0.434	0.645	0.777	< 0.001
Floral	0.758	-0.113	0.767	< 0.001
Balsamic	0.246	0.707	0.749	< 0.001
White Floral	0.702	-0.136	0.715	< 0.001
Whiskey	-0.333	0.627	0.710	< 0.001
Anis	-0.497	-0.477	0.689	< 0.001
Conifer	-0.050	0.687	0.688	< 0.001
Mineral	-0.411	0.551	0.687	< 0.001
Tobacco	-0.524	0.426	0.676	< 0.001
Citrus	-0.154	-0.644	0.662	< 0.001
Sour	0.595	-0.207	0.630	< 0.001
Fruity	0.411	-0.458	0.616	< 0.001
Marine	-0.453	-0.413	0.613	< 0.001
Soapy	-0.213	0.562	0.601	< 0.001
Honey	-0.259	0.535	0.595	< 0.001
Chocolate	0.191	0.539	0.572	< 0.001
Leather	-0.472	-0.159	0.498	< 0.001
Cacao	-0.088	0.473	0.481	0.003
Aromatic	-0.479	-0.038	0.480	< 0.001
Warm Spicy	-0.383	-0.283	0.477	< 0.001
Fresh Spicy	-0.431	-0.174	0.465	< 0.001
Aquatic	-0.008	-0.454	0.454	< 0.001
Woody	-0.323	-0.297	0.439	< 0.001
Salty	-0.018	0.415	0.416	0.026
Smoky	-0.085	0.406	0.415	< 0.001
Coffee	-0.181	0.368	0.410	< 0.001
Rose	0.404	0.041	0.406	< 0.001
Iris	0.218	0.341	0.405	< 0.001
Champagne	0.066	0.370	0.376	0.012
Yellow Floral	0.256	0.264	0.368	< 0.001

Note. Accords are ranked by their Euclidean vector length in the two-dimensional LDA space, $\text{Norm}_k = \sqrt{LD1_k^2 + LD2_k^2}$, which captures the overall contribution of each accord to segment separation. LD1 and LD2 are estimated using only accords significant in at least one equation of the multinomial logit model. The minimum p -value is the smallest across the two multinomial logit equations for each accord.

H Web Appendix H: Balanced Sample Construction

Because the subset of fragrances with reviews differs systematically from the underlying fragrance population, we construct a balanced sample of fragrances using a stratified propensity-based procedure (Valliant, Dever, and Kreuter, 2013) within each gender segment. Specifically, we estimate a propensity score for review availability (equal to 1 if a fragrance has at least one textual review and 0 otherwise) using logistic regression with the full set of fragrance characteristics used in our empirical analysis, including atypicality, average rating, number of ratings, number of accords, brand, decade of release, perfumer identities, and main accords.

Among fragrances with reviews, we assign inverse-propensity weights equal to the inverse of the estimated probability of review availability. Within each gender segment, we then search for the largest weighted random subsample that remains balanced relative to the full fragrance population. Candidate sample sizes are evaluated in increments of 100, starting from the largest feasible size and proceeding downward. For each candidate size, we assess balance on atypicality, average rating, number of ratings, and number of accords using two-sided Welch tests. The selected sample is the largest one for which equality of means cannot be rejected for all four variables at the 5% level.

After balancing, the sample contains 4,500 men’s fragrances, 9,600 women’s fragrances, and 7,100 unisex fragrances. Table H.1 compares the full fragrance population, the reviewed sample, and the balanced sample. Whereas the reviewed sample differs significantly from the underlying population on several fragrance characteristics, the balanced sample closely reproduces the corresponding population distributions across all three segments. Differences in means are uniformly small, and none of the balance tests reject equality of means at the 5% level.

Table H.1: Covariate Balance Across Population, Reviewed, and Balanced Samples

Segment	Variable	Population	Reviewed Sample			Balanced Sample		
		Avg.	Avg.	Diff.	<i>p</i>	Avg.	Diff.	<i>p</i>
Men	Atypicality	0.316	0.313	-0.003	0.233	0.316	-0.001	0.803
	Number of Ratings	294.794	401.027	106.233	< 0.001	333.124	38.330	0.058
	Rating	3.979	4.004	0.025	< 0.001	3.987	0.008	0.269
	Number of Accords	8.491	8.496	0.005	0.869	8.454	-0.038	0.300
	Sample size	10,786	7,474			4,500		
Women	Atypicality	0.396	0.395	-0.001	0.522	0.396	0.000	0.891
	Number of Ratings	284.906	399.749	114.843	< 0.001	307.327	22.421	0.066
	Rating	3.948	3.947	-0.001	0.873	3.948	-0.000	0.934
	Number of Accords	8.053	8.150	0.097	< 0.001	8.057	0.003	0.905
	Sample size	25,468	17,556			9,600		
Unisex	Atypicality	0.415	0.415	0.000	0.982	0.415	-0.000	0.850
	Number of Ratings	150.289	224.481	74.192	< 0.001	162.983	12.694	0.088
	Rating	4.006	3.993	-0.014	0.004	3.998	-0.008	0.182
	Number of Accords	7.934	7.920	-0.014	0.552	7.903	-0.032	0.332
	Sample size	25,614	16,588			7,100		

Note. The table compares the full fragrance population, the reviewed sample, and the balanced sample. Entries report means, differences relative to the full population, and two-sided Welch test *p*-values. Sample sizes were selected separately within each gender segment as the largest balanced subsamples for which equality of means with the corresponding population could not be rejected at the 5% level for atypicality, average rating, number of ratings, and number of accords.

I Web Appendix I: Discriminative Theme Extraction

We provide additional details on the procedure used to identify the themes that distinguish typical and atypical fragrances within each target segment.

The procedure combines supervised discrimination with unsupervised clustering at the sentence level. Reviews from the focal groups (Q_1 and Q_4) are first segmented into sentences. We retain only English sentences whose lengths fall between the 10th and 90th percentiles of the sentence-length distribution and remove exact duplicates. Each sentence is embedded using the pretrained BAAI/bge-large-en-v1.5 model. A random forest classifier then assigns each sentence a discriminative score reflecting its association with the focal group. We retain the top 10% most discriminative sentences and cluster their embeddings using k -means with cosine distance, selecting the number of clusters according to the silhouette criterion. Algorithm 1 summarizes the complete procedure.

Algorithm 1 Discriminative Theme Extraction

Require: Reviews $\mathcal{R}_{Q_1}, \mathcal{R}_{Q_4}$; candidate grid $\mathcal{K}$

Ensure: Themes $\{\mathcal{C}_c, \mathcal{T}_c\}_{c=1}^{k^*}$

```

1:  $\mathcal{S} = \{(s_i, y_i)\}_{i=1}^N \leftarrow \text{PREPROCESS}(\mathcal{R}_{Q_1}, \mathcal{R}_{Q_4})$  ▷ Split reviews into sentences
2:  $\mathbf{e}_i \leftarrow f_{\text{BGE}}(s_i)$  ▷ Sentence embeddings
3:  $\hat{p}_i \leftarrow \Pr(y_i = 1 \mid \mathbf{e}_i)$  ▷ Random forest discrimination
4:  $\mathcal{S}^* \leftarrow \{s_i : \hat{p}_i \geq q_{0.90}(\hat{p})\}$  ▷ Top 10% most diagnostic sentences
5: for  $k \in \mathcal{K}$  do
6:    $\mathcal{C}^{(k)} \leftarrow \text{KMEANS}_{\text{cos}}(\mathcal{S}^*, k)$ 
7:   Compute Sil( $k$ ) ▷ Average silhouette
8: end for
9:  $k^* \leftarrow \arg \max_{k \in \mathcal{K}} \text{Sil}(k)$  ▷ Select number of themes
10:  $\mathcal{C}_1, \dots, \mathcal{C}_{k^*} \leftarrow \mathcal{C}^{(k^*)}$ 
11: for  $c = 1, \dots, k^*$  do
12:    $\mathcal{T}_c \leftarrow \text{Nearest}_8(\cos(\mathbf{e}_i, \bar{\mathbf{e}}_c))$  ▷ Representative sentences
13: end for
14: return  $\{\mathcal{C}_c, \mathcal{T}_c\}_{c=1}^{k^*}$ 

```

Theme discovery was performed separately for men’s, women’s, and unisex fragrances by applying k -means clustering to the top 10% most discriminative sentences. Candidate solutions with $k \in \{10, \dots, 20\}$ were evaluated using the silhouette coefficient. Table I.1 summarizes the clustering diagnostics. The optimal solutions were $k = 13$ for men’s fragrances (silhouette = 0.0243), $k = 17$ for women’s fragrances (0.0201), and $k = 15$ for unisex fragrances (0.0301). The resulting clusters ranged in size from 319 to 1,475 sentences for men, 277 to 2,383 for women, and 305 to 1,490 for unisex fragrances.

Table I.1: Cluster Selection Results for Theme Discovery

Segment	k	Sentences	Inertia	Silhouette	Min. Size	Max. Size
Men	10	12,968	5748.657	0.022	486	1,922
	11	12,968	5711.465	0.023	467	1,868
	12	12,968	5677.051	0.023	458	1,541
	13	12,968	5646.262	0.024	319	1,475
	14	12,968	5620.398	0.021	307	1,259
	15	12,968	5597.183	0.021	323	1,226
	16	12,968	5572.310	0.023	310	1,238
	17	12,968	5552.653	0.024	291	1,213
	18	12,968	5531.994	0.022	316	1,054
	19	12,968	5514.270	0.023	310	1,037
	20	12,968	5493.802	0.024	308	1,066
Women	10	23,631	10839.555	0.017	929	3,102
	11	23,631	10785.031	0.017	914	3,043
	12	23,631	10732.287	0.017	924	2,820
	13	23,631	10687.365	0.017	901	2,767
	14	23,631	10641.790	0.015	910	2,227
	15	23,631	10594.254	0.019	278	2,558
	16	23,631	10557.344	0.017	743	2,134
	17	23,631	10519.261	0.020	277	2,383
	18	23,631	10477.409	0.020	280	2,208
	19	23,631	10441.878	0.020	275	2,055
	20	23,631	10409.284	0.019	276	1,783
Unisex	10	11,687	4991.015	0.028	462	1,844
	11	11,687	4962.934	0.028	468	1,753
	12	11,687	4936.805	0.029	434	1,661
	13	11,687	4911.908	0.029	393	1,658
	14	11,687	4889.721	0.030	420	1,538
	15	11,687	4868.180	0.030	305	1,490
	16	11,687	4849.323	0.030	306	1,479
	17	11,687	4830.799	0.020	341	1,136
	18	11,687	4813.281	0.021	346	999
	19	11,687	4791.997	0.025	339	1,337
	20	11,687	4774.120	0.021	314	897

J Web Appendix J: Themes and Representative Expressions (Exploratory Sample)

Tables J.1–J.10 report the discovered themes and representative expressions identified in the exploratory sample for atypical (Q_4) men’s and women’s fragrances and typical (Q_1) unisex fragrances.

Table J.1: Themes Distinguishing Atypical from Typical Men’s Fragrances (Exploratory Sample)

Theme M1: Seasonal Suitability and Cold-Weather Appeal
- This scent is suitable for the fall and winter. - It’s a cold weather fragrance. - It’s preferably a cold weather or nighttime scent. - Occasions: I think this is a winter scent. - Its a great winter or cold day in the fall scent. - Suitable for cold weather - it evokes warmth, change of seasons, sophistication. - This is for cold nights. - Certainly cold weather/winter/fall scent. - Smells like it would be a great fall/winter fragrance. - A very nice scent for the fall and winter.
Theme M2: Material-Centric Recombination (Oud)
- There’s some oud in the mix. - I also get a hint of oud. - There is quite some oud lingering through. - Something like synthetic oud in the mix too. - And a bit of oud. - if you like oud, try it. - Can anyone else smell oud in this? - The Oud scent is pleasant and warm. - MIXED WITH OUD AND LEATHER. - One of the best oud scents out there.
Theme M3: Intense Positive Evaluation and Aesthetic Appeal
- This is a spectacular scent. - This is an exquisite scent. - This is a glorious scent. - This scent is magnificent. - This is a very interesting scent. - This is an amazing smelling scent. - This is a beautiful scent. - The scent is amazing. - This scent is intoxicating and alluring. - This is a really beautiful scent.
Theme M4: Material-Centric Recombination (Leather)
- A nice leather scent. - Leather, leather and a bit more leather is what I get from this fragrance. - Lovely Scent of Leather. - The leather is dominant. - Here leather is utilized as an accent. - This has leather that makes it not too fruity. - It’s a rich leather, and to me has a slight smoky and fruity quality. - This is what leather is supposed to smell like. - This is a warm, spicy leather fragrance. - A beautiful leather and aromatic chypre.

Table J.2: Themes Distinguishing Atypical from Typical Men's Fragrances (Exploratory Sample)

Theme M5: Ownership, Sampling, and Acquisition Experiences
- Ps: I have one of the first batches might be less nuclear in current batches (200ml bottle) - Bought a decant to sample. - I really like this a lot and am tempted to buy a bottle, I would suggest you get a decant and test it a good week. - Decided to get a sample of this instead of just buying it. - I have a 5ml sample of this (unsure if it's the EDT or EDP, however it says ESSENZE like this one so I'm leaving my review here). - So I bought a sample. - First impression, sweet caramel, warm, cozy for the cold weather deffo. I can see bold women in a leather jacket pulling this off too. 58€ for 125ml. - You can smell the same almond/cherry note in the original L'homme Ideal that's widely available to test so I would advise doing that first. - Fragx and Fragnet both have 60ml bottles for around 70–75... So, solid pickup... Had to do it. - Prezzo 30 ml for 30 € ... but the cinnamon COVERS oud ... and frankly I want to feel a stinging oud not a sweet oud.
Theme M6: Hedonic Richness and Emotional Appeal
- Got a sample and this is absolutely breathtaking. - This stuff is gorgeous. - I'm in awe of how much I like this - This is becoming one of my favorites. - It is nothing shy of intoxicating. - Damn, I love this stuff. - This is so damn good. - This is gorgeous and sexy. - Oh my GOODNESS, this is so heavenly. - It's just so good.
Theme M7: Cross-Gender Recombination (Sweet and Gourmand Accords)
- There is a sweetness in there as well. - It's slightly sweet and musky. - There's a subtle sweetness in there, too. - The vanilla adds a sweet touch. - There is a sweet/gourmand aspect here too. - There's more of a creamy sweetness to it than anything else. - It is sweet but not in a cloying way. - It has vanilla which makes it sweeter. - This is very fruity and sweet. - There's also a bit of sweetness up top as well.
Theme M8: Intensity and Sensory Presence
- It is VERY bold. - This is a strong like. - It's subtle and strong. - Well, this has shades of that, and I'm not kidding. - Interesting vibes I get from this. - The color promises a lot. - It's also a little bit earthy. - It's FIERCE, but Fiercer. - There's something very sexy going on here. - This is like the intense version.
Theme M9: Idiosyncratic Olfactory Impressions and Sensory Ambiguity
- To my nose, this is almost a skin scent. - It smells like someone has an interesting scent. - It is a pungent, almost semi-sweet smell. - It's fruity, almost fruit loop type of smell. - The lingering scent is crazy. - It's kind of hard to describe the scent well. - This is how it smells to me. - After that it's a (fairly strong) skin scent. - This one is a bareble animalic scent . - Very weird scent comes off of it.

Table J.3: Themes Distinguishing Atypical from Typical Men’s Fragrances (Exploratory Sample)

Theme M10: Negative Expectation Disconfirmation and Taste-Based Rejection
- This is not, in my opinion, a casual scent. - A kinda interesting scent for a beginner scent, but not for me. - I really thought I would like this fragrance, but unfortunately it doesn’t sit well with me. - I’m not much into leathery scents and this could be a reason for my dislike. - Not a bad fragrance but just not my cup of tea. - I like sweet fragrances but something about the sweetness in this one made me feel I’ll, not a fragrance for me. - However, I do think this scent is not something that everyone would like. - This fragrance is a put off for me! - It’s not really a skin scent. - This is not a perfume for the many.
Theme M11: Gender Boundary Blurring (Unisex Perception)
- The fragrance is not overly masculine, rather it is more unisex IMO. - This is Unisex fragrance. - It is listed as a unisex fragrance, but personally I find it very masculine. - I would call this a unisex fragrance. - This a unisex fragrance. - I’m female, and this scent is quite unisex. - This is unisex maybe even a little on the feminine side. - is a unisex scent in my opinion. - I consider this scent unisex, not just for men. - It is supposed to be unisex, but comes off somewhat slightly more feminine to my nose (but that seems the trend for masculine scents recently).
Theme M12: Within-Category Recombination (Tobacco and Smoky Accords)
- Giving the tobacco a sweet and smoky vibe. - It’s aromatic, almost delicious. - The mix of smoke, sweet and leather is very attractive. - It’s a nice tobacco scent and almost has a patchouli like quality to it. - The vanilla gives it almost a woody quality. - Resinous and almost earthy feeling with a hint of aromatic nuances. - Masculine, ambery, with a touch of sweetness and smoke. - I notice some slight incense or maybe that’s just the tobacco. - Its got some spice. - Slight hint of tobacco and rum is noticeable.
Theme M13: Ambivalence and Taste-Based Uncertainty
- But this is not for me. - Its not for me. - It is not for everyone. - It is not for everybody. - However, there’s something in it that doesn’t sit well with me personally. - Unfortunately, it’s not something everyone will like. - It’s not for everyone. - I have very mixed feelings in this frag. - I’m not sure how I feel about this one. - It’s not entirely friendly.

Table J.4: Themes Distinguishing Atypical from Typical Women's Fragrances (Exploratory Sample)

Theme W1: Evocative Depth and Atmospheric Complexity
- There is something slightly smoky, airy about it. - There is also a hint of leather in there. - It has a slight smokiness in it. - And a hint of something leathery. - that is reminiscent of leather. - It's soft, and slightly powdery. - There's something very green in it. - It is heavier, more lush. - This has something of that in it. - This has a plastic, waxy nuance to it.
Theme W2: Within-Category Recombination (Rose Intensification)
- This is a rose perfume. - The heart of the fragrances is rose. - One of the most beautiful rose fragrances imo. - One of my favourite scents with rose in it. - It's a romantic and screeching rose. - This is THE rose perfume. - This is definitely a rose scent. - Smells of an antique rose. - A beautiful true rose scent that lingers and mystifies. - This perfume is for the "rose lovers".
Theme W3: Negative Expectation Disconfirmation (Category Ambiguity)
- It's not sticky sweet like usual gourmand scents. - It's not quite a skin scent, but not very far from it. - As others have mentioned, this is not a musk scent. - More of a skin scent than a perfume. - It is not a warm or cold scent. - Not as resinous as you would think. - It is not what I'd call extremely gourmand though, and it is not cloying. - I was expecting less of a gourmand fragrance with this one but that's not the case here. - It is not a warm scent, nor a gourmand scent. - It is warm but not syrupy.
Theme W4: Within-Category Recombination (Creamy Gourmand Accords)
- It also has a touch of vanilla in it. - It has a lovely creaminess to it. - There is a hint of vanilla to round it all. - This could also have a dash of vanilla. - It's kind of delicious. - it also has a hint of almonddiness. - It's creamy, fleshy, and sweet. - A dash of creamy, sweet, milky-ness. - There is also a warm, soft vanilla that's kind of caramelly. - but there is creaminess to it.

Table J.5: Themes Distinguishing Atypical from Typical Women’s Fragrances (Exploratory Sample)

Theme W5: Within-Category Recombination (Coconut Gourmand Accords)
- This is a pretty coconut scent. - It’s a coconut and vanilla scent. - This is a nutty coconut fragrance. - It smells like coconut and vanilla. - A very nutty/ buttery sweet coconut scent. - A floral coconut fragrance. - And a small hint of coconut. - smells like frosted coconut. - This smells like coconut cookies to me . - This smells like a creamy coconut with sugar.
Theme W6: Cross-Gender Recombination (Spicy and Resinous Accords)
- It’s herbal and aromatic. - I like how there’s a tinge of incense in there as well. - There is some spice, some jasmine, some sweetness. - This scent is rich and complex. - A unique gourmand scent. - It has a little bit of incense too. - The spices also come through and make the scent even more interesting. - I think there’s some amber, incense and soft floral notes in there as well. - Curious blend that creates a somewhat exotic fragrance. - The scent is spicy, sweet, and woody.
Theme W7: Novelty and Distinctiveness
- There is an incensey feel to the scent. - This scent is interesting. - this is an interesting scent. - This is an interesting scent. - Reminds me of the smell of a Lush shop. - It almost smells a hint of Creamsicle. . - The smell of this seems to me to be very close to nature. - This is what it smells like. - THIS IS WHAT IT SMELLS LIKE. - Oddly what I’m now smelling is something very, very lovely.
Theme W8: Performance Intensity and Wearability
- It lasts a long time on skin. - It lasts for a long time. - The scent lasts and lasts. - It lasts a long time. - It smells quite strong but not overpowering, it lasts long too. - On me this is a skin scent that lasts several hours. - And it lasts long. - For a body mist, this fragrance has many hours of longevity. - It lasts and it is strong but not headachy. - It is extremely fragrant and long lasting.

Table J.6: Themes Distinguishing Atypical from Typical Women’s Fragrances (Exploratory Sample)

Theme W9: Gender Boundary Blurring (Unisex Perception)
- This fragrance is UNISEX. - It is very unisex. - Definitely a unisex type of scent. - Definitely a unisex scent. - It is def a unisex fragrance in my opinion. - There is slightly unisex vibe. - A lovely fragrance that is definitely unisex. - It could be unisex scent. - It's unisex in my opinion. - Undoubtedly a unisex fragrance.
Theme W10: Within-Category Recombination (Floral Intensification)
- This fragrance is for tuberose lovers. - this is a beautiful tuberose scent. - This is basically a tuberose scent, but a nice tuberose. - This perfume makes me love tuberose. - A beautifully blended scent with a focus on tuberose. - Naturally smelling tuberose with a hint of jasmine. - i feel a scent of tuberose in it. - For tuberose lovers, this could be perfect scent. - It is a tuberose centric perfume. - A lush, dense tuberose/gardenia scent.
Theme W11: Perceived Artistry and Refined Appeal
- The artistry is exquisite. - A work of art. - It is absolutely lovely. - It is really gorgeous. - It's a real beauty. - There is certainly something mystical in this one. - Something about this is powerful and sophisticated. - The quality is notable. - The quality is stunning. - There is something very special about this.
Theme W12: Intense Positive Affect and Enthusiastic Evaluation
- Makes me feel delicious, and I seriously dig it. - I love this, I covet it. - So, I just received a sample of this and am swooning. - I just adore this. - I believe I have finally stumbled upon my favourite Amouage. - For me this is lovely. - God this smells so good on me ! - This is absolutely SCRUMPTIOUS. - I cannot wait to sample this. - I enjoy this a lot!
Theme W13: Within-Category Recombination (White Floral Intensification)
- A delightful white floral. - A great floral chypre. - A lovely chypre flower. - This is a lovely white floral. - There's clove bud /carnation. - It's a gourmand floral. - It smells of pure flowers. - This is an exotic floral blast. - Some white florals can be intoxicating in that way. - Beautiful yet simple white flora.

Table J.7: Themes Distinguishing Atypical from Typical Women’s Fragrances (Exploratory Sample)

Theme W14: Evocative Sensory Appeal and Hedonic Richness
- This scent is delicious. - This scent is delectable. - This fragrance is sublime. - It’s a delicious scent. - This is the beauty of this scent. - This scent is quite admirable. - This is an extraordinary scent. - This is a fantastic scent. - This scent is so yummy. - This fragrance is a wow-zer.
Theme W15: Accord-Profile Disconfirmation
- am a fan of sweet coconut scents, however this is not for me. - It smells herbal, kind of like dried oregano and thyme :/ - As someone who prefers balsamic, smooth/rounded scents with many of the listed notes, this is simply not my cuppa. - I know Vanilla isn’t listed, it must be the Amber I’m picking up; but my nose reads it like a really warm, completely NOT sweet vanilla. - To my nose this is more akin to an oriental spicy than a floral. - Perhaps benzoin is at the base that is giving me vanilla vibes, as opposed to amber, but my nose is not entirely sure. - Sandalwood is an iffy fragrance on me and in this case it ruins my appreciation of a nicely blended iris fragrance. - Generally, I appreciate a nice gourmand vanilla with some sweetness in it, however, this one was too much for my nose. - I’m not one for coffee scents, probably ever, but I can’t get over this one. - Smells almost salty and medicinal and I can’t quite get the floral notes that are supposedly in here.
Theme W16: Ownership, Sampling, and Comparative Experiences
- Reminds me of Sensuous Nude by Estée Lauder a bit, it’s also sweet vanilla coconut perfum, it was heaven. - Ask for a small 5ml bottle at the shop to decide if this fragrance is yours ;) - the body lotion “rosé” smells identical to idole by lancome, i havent used the parfum but it seems logical to me that they have similar fragrances - Edit: layering this with Gourmand Coquin gives off Feve Delicieuse vibes. - I layered Diptique Eau Duelle recently and it was heavenly on top of this, just gives it an extra kick of herbal vanilla. - EDIT: I ended up falling in love with this scent and just purchased a full bottle! - If anybody knows of something that smells almost just like this but better quality please pm me :) - UPDATE: I just received my decant, and what an exquisite fragrance this is! - Edit: I tried layering it with Acqua di Colonia by Santa Maria Novella, and surprisingly the smell of this perfume stood out as more creamy and sweet! - I am looking for a salty rose fragrance and would love to test this one in particular.
Theme W17: Ambivalence and Qualified Evaluation
- I have mixed feelings on this one. - This is one to try before buying. - Might have to experiment. - It’s not spot on. - I suggest sampling this before you buy if you can. - Not thrilled by this, unfortunately. - Not a straight dupe. - Please try before buying. - I had a sample of this. - This is the latter kind.

Table J.8: Themes Distinguishing Typical from Atypical Unisex Fragrances (Exploratory Sample)

Theme U1: Wearability and Experiential Appeal
- Makes for a more intimate wearing. - Can be worn in most occasions. - It compliments it actually. - I have gotten compliments. - This is for cooler weather. - The more I wear it, the more I like it. - It's warm and interesting. - The more I wear it the more I like it. - I ended up liking it more in subsequent wearings. - It's an interesting mix that garners compliments and can be worn in the mildly warmer months.
Theme U2: Balanced Olfactory Recombination (Warm and Resinous Accords)
- i like the hint of sandalwood. - A touch of something herbal and some woods thrown in as well. - It is a beautiful fruity scent with some herbs. - Beautiful incense fragrance with a slight sweet tinge. - Has some light vanilla along with bursts of yummy sandalwood and cinnamon. - Some airy sandalwood too. - It's a little balsamic, resinous and ambery which adds depth, warmth and some lovely sweetness. - Amber and vanilla rounds it out and makes this a cozy scent. - Warm cinnamon and gentle sandalwood. - It's an interesting mix of sweetness, spice, and wood.
Theme U3: Ownership, Sampling, and Value-Based Evaluation
- Personally I wouldn't buy it if it cost € 10,00 but if you're tempted to, at the absurd retail price, please, sample first and not just once. - Price is high but for a niche one, it's "okay-ish". - Still, at \$140 for 50ml, the price is a bit steep, and thus would strongly recommend not to blind buy that one. - I do not think it is blind buy worthy at this price point. - Most fragrances are not \$220 [for 50ml] worthy (quite ridiculous actually), but some are \$300 [for 100ml] worthy though :) - If you like this aroma and want to pay for a product itself then look for cheaper options, or indie brands if you like to spend more. - Tbh, similar to the VC&A, if I found this for half-price I'd buy it but def not for full price. - Had to try this for the price and the ranking. - I got a sample of this and tried it before looking at the price. - I do think a full bottle is overpriced for what it is though (at 120-150\$) in the US.
Theme U4: Balanced Olfactory Recombination (Woody and Aromatic Accords)
- This is a woody and spicy fragrance. - This is a very aromatic woody fragrance. - woody smokey/incense-y with a hint of sweetness. - It's somewhat aromatic also. - It has a bit of a woody scent with a touch of vanilla. - A woody scent with polished furniture feel. - There's spice and incense and wood and pine in there too. - It is a scent with an unusual combination of accords. - To me, this is a very aromatic woody and greenish scent. - I can smell a touch of herb and spice on the background.
Theme U5: Perceived Artistry and Distinctive Appeal
- This is a beautiful fragrance. - This is a fragrance with personality. - This fragrance is beautiful. - This fragrance is BEAUTIFUL. - This is a wonderful fragrance. - It's a great fragrance. - This fragrance is interesting. - This is a great artisanal fragrance . - This fragrance is well done. - This perfume is lovely.

Table J.9: Themes Distinguishing Typical from Atypical Unisex Fragrances (Exploratory Sample)

Theme U6: Qualified Evaluation and Mild Disappointment
- Different, but doesn't wow me. - I like this, but I don't love it. - It is very distinctive, but not in a good way. - The performance isn't great either. - Doesn't impress me very much. - It's not terribly unique. - A subjective opinion about this. - Doesn't project well and isn't entirely that unique. - Overall I'm not immensely impressed but that's probably due to my taste. - For me, this misses the mark.
Theme U7: Perceived Artistry and Compositional Excellence
- this is a work of art. - This is a work of art. - The performance is stellar. - THIS is a beauty. - This is a beauty. - This is an astounding composition. - The quality and blend are superb. - The notes infuse extremely well with each other. - It's really good and unique. - Crisp, clean, natural, invigorating, light and yet it performs.
Theme U8: Gender Perception (Directional Lean)
- Unisex, but leans a bit masculine. - Unisex, but leans masculine. - Leaning a little masculine but seems unisex. - Unisex, but leans more towards masculine I would say. - More masculine than unisex IMO. - Unisex but leans masculine. - I would call this a unisex, though it does lean toward masculine. - Unisex although it leans slightly feminine. - Unisex but definitely leans a bit feminine. - It's unisex but leans feminine.
Theme U9: Balanced Olfactory Recombination (Oud Integration)
- You get the oud mixed with floral notes. - Oud is strong with this one. - Very versatile to be an oud fragrance. - This is a very pleasant oud fragrance. - I like this better than oud for greatness, it's more of a fresh oud scent with an old school feeling about it. - Another Amouroud "Oud" that doesn't smell like oud. - This is Oud done well. - Sometimes OUD can be overwhelming but in this fragrance it is blended the right way. - The oud has a musky animalic touch. - This is not your typical skanky pungent Oud.
Theme U10: Negative Expectation Disconfirmation (Conceptual Ambiguity)
- This fragrance is not a masterpiece. - For me there is a bit of a disconnect between the name of this fragrance and the scent of this fragrance. - An artisan fragrance as opposed to a utility one, perhaps. - Good fragrance but not great. - Perhaps this is not a girly fragrance. - It's different from most fragrances out there. - This fragrance isn't as spicy or as oriental as I expected it to be. - There are better smelling fragrances out there with more of a technical backbone. - To me this fragrance doesn't know what it wants to be. - I have some different things on this fragrance.

Table J.10: Themes Distinguishing Typical from Atypical Unisex Fragrances (Exploratory Sample)

Theme U11: Balanced Olfactory Recombination (Temporal Evolution) - It dries down to a floral/spicey/woody base. - There is a sweet woody amber accord that gets more noticeable in the drydown. - As it dries, this fragrance begins to get a woodiness to it. - The woodiness on the drydown is delish. - Dries down to a vanilic woody scent. - As the fragrance dries down, it becomes a little darker and woodsier. - You can smell the vanilla and amber more in the drydown. - In the final dry down there's something incense like. - Kinda reminds me a bit of the woody dry down on some of the l'interdit perfume by givenchy. - A little musky too, but the drydown has a lovely mild spice in it.
Theme U12: Balanced Olfactory Recombination (Layered Integration) - Opens up with green and citrusy notes. - There is a citrus start, but then the spice kicks in. - Start is a citrusy blast with a hint of spice. - There's also a touch of sharp citrus in the opening combined with some spices. - Starts off with a citrus note. - There's a bit of spice emerging as well. - I get a nice spice and a nice floral and some citrus and some woods. - The citrus notes give it some sparkle. - The opening is really woody, peppery with citrus to me. - This gives me a blast of lemon at first, then it becomes complex with fresh spicy, aromatic/herbal notes.
Theme U13: Performance Intensity and Wearability - Performance is good but not extreme, 7-8 hours longevity and good projection for 2-3 hours. - Longevity is about 6-8 hours, with excellent projection for 2-2.5 hours. - Moderate projection for 2-3hr, longevity 6-8hr. - Longevity is 7-8 hours. - Longevity is over 8 hours, sillage is small to medium. - Longevity is good 8 hours. - Performance is good, not a beast but I get about 5-6 hours longevity with good projection and good sillage. - 8h longevity, gentle projection. - Projection is decent, but longevity is a bit shorter than I'd expect... maybe 5 6 hours tops. - longevity is about 8-10 hours and projection is medium.
Theme U14: Comparative Evaluation (Scent Similarity) - It reminds me a lot of Eau Duelle Diptyque edp in this way. - Kind of similar with Chanel Eau Tendre. - Reminds me a bit of Kalemata except Kalemata smells smoother. - Haven't yet smelled another fragrance that is too similar to this. - Reminds me of many fragrances too but still has it's own thing going on. - Though I may say this is not an exact dupe, this smells very close to it, I would say 75-80 percent alike. - Almost in a cologne-like way. - A fragrance which is similar to others I have tried from initial to dry down phase. - I'm getting something very similar to Tom Ford Black Orchid from this, maybe closer to the sweeter Parfum version. - There is also something in this composition which reminds me of Mancera Aoud Lemon Mint.
Theme U15: Ownership, Sampling, and Comparative Experiences - I will some day get Creed Spice and Wood, that's a classy fragrance that has a nice freshness, that would justify more the price. - This smells great and I'm hoping that it lasts longer than my ADP (except Essenza which is amazing), 4711 and Penhaligons (Castile) iterations. - Edit add: I tried out Alexandria Fragrances Sublime clone scent called Splendid Vanille. - Update 08/11/21: So I've worn through my sample and I just want to add some details on the scent itself. - UPDATE: I eventually bought this lovely fragrance (after getting Chimaera 2018 Anniversary Edition) and I've been enjoying it. - you can find that for a cheaper Price then 120€. It smells like you Mix Maison Martin Margiela-Replica with Acqua di Parma -Mandorlo. - LOL Sublime Vanille is clean, soft, smells expensive (but light) and I think would be great on guys and girls. - Scent 9/10 Longevity 8.5/10 Sillage 8.5/10 Price worthy 7.5/10 - To me, it smells similar to Maison Francis Kurkdjian - Oud Satin Mood...maybe a slightly lighter version...now, which one do i like better? - ETA: Eau de Sisley No3 could teach Karma a thing or two about a great citrus/patchouli combo.

K Web Appendix K: Confirmatory Validation of Themes

Table [K.1](#) compares the prevalence of the exploratory themes in the exploratory and confirmatory samples. Across all three fragrance segments, the distributions are stable, with only small differences in prevalence for individual themes. These results indicate that the themes identified in the exploratory sample generalize well to the confirmatory sample and are not driven by idiosyncrasies of the exploratory review data.

Tables [K.2–K.11](#) report representative sentences for each theme from the confirmatory review data. The reported examples are obtained by projecting the sentences of the confirmatory reviews into the same embedding space used for exploratory theme discovery. Within each theme, confirmatory sentences are ranked by their cosine similarity to the corresponding exploratory cluster centroid and the ten nearest unique sentences are reported.

Table K.1: Theme Distribution Stability Between Exploratory and Confirmatory Samples

Segment	Theme	Exploratory (%)	Confirmatory (%)	Difference (%)
Men	Other Themes	90.000	89.160	-0.839
	M1	0.246	0.289	0.043
	M2	0.450	0.461	0.011
	M3	1.137	1.635	0.498
	M4	0.742	0.844	0.102
	M5	0.777	0.865	0.088
	M6	0.773	0.768	-0.005
	M7	0.908	0.960	0.052
	M8	0.977	0.995	0.018
	M9	1.002	1.095	0.093
	M10	0.947	0.887	-0.060
	M11	0.345	0.436	0.090
	M12	0.924	0.818	-0.106
	M13	0.772	0.787	0.015
Women	Other Themes	90.000	91.149	1.149
	W1	0.699	0.555	-0.143
	W2	0.354	0.298	-0.055
	W3	0.534	0.412	-0.123
	W4	0.857	0.734	-0.122
	W5	0.324	0.290	-0.034
	W6	0.830	0.615	-0.215
	W7	0.895	1.006	0.110
	W8	0.344	0.365	0.021
	W9	0.117	0.165	0.048
	W10	0.371	0.368	-0.003
	W11	0.613	0.524	-0.088
	W12	0.648	0.611	-0.037
	W13	0.561	0.473	-0.088
	W14	1.008	1.033	0.024
	W15	0.732	0.497	-0.235
	W16	0.660	0.490	-0.170
	W17	0.453	0.415	-0.039
Unisex	Other Themes	90.000	88.135	-1.865
	U1	0.531	0.552	0.022
	U2	0.880	0.961	0.081
	U3	0.400	0.427	0.026
	U4	0.840	1.127	0.287
	U5	1.275	1.981	0.706
	U6	0.815	0.738	-0.077
	U7	0.888	0.964	0.075
	U8	0.624	0.834	0.210
	U9	0.349	0.412	0.063
	U10	0.835	0.973	0.138
	U11	0.471	0.539	0.067
	U12	0.565	0.703	0.138
	U13	0.261	0.342	0.081
	U14	0.589	0.685	0.097
	U15	0.678	0.628	-0.050

Table K.2: Themes Distinguishing Atypical from Typical Men's Fragrances (Confirmatory Sample)

Theme M1: Seasonal Suitability and Cold-Weather Appeal
- This is a night out fall/winter fragrance. - Cold weather fragrance when dressed for the occasion. - This is a lovely fragrance for cold weather and especially evenings out. - It can be used at cold winter nights. - It is a heavy fragrance mostly for winter. - This one is suitable for winter, fall and colder nights mostly. - This fragrance is more suited to be used at night, colder weathers. - I prefer to wear this scent in fall and the winter. - Definitely a winter/fall night fragrance. - A nice one for the colder months.
Theme M2: Material-Centric Recombination (Oud)
- There is something powdery Vanilla/ Coffee/Patchouli/Cashmere about this Oud. - Nice scent The spices are wrapped around the oud. - this one is a dry/spicy oud Its a nice fragrance that oud, woody notes, cinnamon, vetiver and pink pepper are present. - This is an amazing oud fragrance. - Oud is also rich and has a bite to it. - Oud in this blend is PRESENT but its not like you may expect. - A great alternative for oud lovers. - The oud is there, but it's synthetic oud. - This is what Leather Oud is. - Maybe it's the oud or something deep and woody.
Theme M3: Intense Positive Evaluation and Aesthetic Appeal
- This is a lovely scent. - The scent is amazing. - It's a spectacular fragrance. - It is a stunning scent. - This scent is magnificent - It is a sublime fragrance. - This is a special scent. - This is a majestic scent. - This is VERY perfumey. - This scent is so unique.
Theme M4: Material-Centric Recombination (Leather)
- Leather centric fragrance with hints of old school. - This is SMOKED leather with a hint of sweetness. - A slightly different and possibly better rendition of leather. - A smoky, resinous leather with a hint of sweetness. - This is a sweet and smokey, leather fragrance. - I believe this is because of the leather. - A classy dark leather based fragrance. - Far more leather, far deeper of a scent as well. - The leather is a noticeable note, with a woody-resinous touch. - The leather is not overpowering.
Theme M5: Ownership, Sampling, and Acquisition Experiences
- Just grabbed a bottle off Muglers site...looking forward to sniffing it..its right up my alley so Im sure it wont be a let down (fingers crossed). - Now its reformulated... a tenth...maybe less... If you can get a vintage sample try it its crazy really crazy :) - Update: Got a sample. - Bought a sample. - Received a sample, and trying it out right now. - I got this as a sample with a recent purchase. - i got a sample and might buy it(i am female). - I have a 100ml bottle if anyone is interested send me a message to negotiate a swap exchange - Just received a sample of this. - Just got a sample of this.

Table K.3: Themes Distinguishing Atypical from Typical Men’s Fragrances (Confirmatory Sample)

Theme M6: Hedonic Richness and Emotional Appeal
- It is almost intoxicating. - The stuff is gorgeous. - It blows my mind how good this is. - This is fabulous stuff. - This is just heavenly stuff. - Man, this is so good. - fuck this is so good. - I received a sample of this and I have been blown away. - In all seriousness, this is amazing. - This may become one of my favourites.
Theme M7: Cross-Gender Recombination (Sweet and Gourmand Accords)
- And there is a bit of sweetness to it. - The sweetness is almost gooey. - There are hints of sweetness in here as well. - There's a light sweetness to it as well. - It's sweet with noticeable vanilla and spices. - It has a sweetness too somewhere in there. - It's supposed to be very sweet (some say too sweet). - There's a touch of sweetness and obviously some mild musk. - It's floral and sweet. - The fragrance is very sweet; some people could find it overly sweet.
Theme M8: Intensity and Sensory Presence
- It is fresh and dark at the same time. - it has its own character to it. - It is also bold. - It has its own character. - After 30min it gets a little bit more masculine. - So it feels different and bold. - This is basically an intense version. - Despite it being a beast, there is a refinement. - A very strong like. - It is reminiscent of amber.
Theme M9: Idiosyncratic Olfactory Impressions and Sensory Ambiguity
- I can smell it. - This smell is quite addictive. - It's a skin scent after an hour or so. - This smells more promising in the air than on my skin at least. - Almost like a kind of menthol like smell. - But a strong skin scent. - After a couple of hours, it's almost a skin scent. - I just figured out what this smells like. - That's what this smell reminds me of. - What this smells like to me (I still have the smell lingering in my nose from 20mins ago lol) is sweet ash.

Table K.4: Themes Distinguishing Atypical from Typical Men’s Fragrances (Confirmatory Sample)

Theme M10: Negative Expectation Disconfirmation and Taste-Based Rejection
- Scent is subjective however this is not for me. - Not my favourite scent. - This isn't a date night scent to me. - The notes of this fragrance do not reflect the smell accurately. - Not a bad fragrance, but not for me. - I think this is not a bad fragrance, just not for me. - This fragrance isn't what I expected looking at the notes. - As a general lover of sweet and woody fragrance, this fragrance does not sit well with my chemistry. - I give this 4 out of 5, as this particular category of scent is not what I prefer. - Just not my kind of fragrance.
Theme M11: Gender Boundary Blurring (Unisex Perception)
- This is a unisex scent IMO. - This is unisex fragrance. - This fragrance is unisex. - I think it's a unisex scent. - This one is a unisex fragrance. - Easily a unisex fragrance IMO. - this perfume can also be unisex. - This is unisex, but more masculine. - As the name suggests, the fragrance is masculine, but I would classify it as unisex. - It's a bit unisex, but leaning towards masculine.
Theme M12: Within-Category Recombination (Tobacco and Smoky Accords)
- There's a subtle whiff of tobacco. - I also smell incense in this, maybe it is the rum and woody notes together. - Either this, or there is a dry tobacco accord in the scent. - Has a light sweetness and a light smokeyness to it. - A layered, delicious, almost edible fragrance. - I also get this tobacco/incense vibe to it, which could be the Tolu balsam and the cedar notes. - Its a woody slightly sweet wet tobacco scent that has a slight smokiness. - A very neat and floral fragrance with a hint of Tobacco or Leather. - The sweet and smokey combo is compelling. - I do agree there is a very slight tobacco vanillish aspect to it, more particularly in the opening when the spices are stronger.
Theme M13: Ambivalence and Taste-Based Uncertainty
- Its not for me. - It's not for me. - But it isn't exactly my taste. - It's not for everyone. - This not for you. - Caution: it is not for everybody. - Not sure how I feel about this one. - this is not a crowd pleaser. - Others may like this, but not for me. - This one is not for me.

Table K.5: Themes Distinguishing Atypical from Typical Women's Fragrances (Confirmatory Sample)

Theme W1: Evocative Depth and Atmospheric Complexity
- Hint of vanilla/patchouli/leather at the end. - There is a slightly waxy greenness. - There is something smokey that adds sophistication. - It's like lacquered or something. - It's kinda spicy at the beginning. - a couple minutes in i get a hint of leather. - There is something a little green and sappy. - There is something woody in the base, a little mature. - It reminds me of Opium. - Sillage is close to skin.
Theme W2: Within-Category Recombination (Rose Intensification)
- A magnificent rose perfume. - It is a jasmine/rose scent. - A very good rose perfume. - On of the best rose scents I've ever tried. - This is a creamy rose. - Nice rose scent with cirtuses and some freesia. - This is a rose that smells aged. - One of the best rose fragrances out there. - Try this if you want to try an interesting rose scent. - This one is a must-try for rose enthusiasts.
Theme W3: Negative Expectation Disconfirmation (Category Ambiguity)
- This is not a spicy scent like others might think. - It's almost more a texture than smell. - It does not smell like grapefruit. - It's more like a "scent" rather than a perfume creation. - There is patchouli, and a fruitiness, but it's not sickeningly sweet. - This is fruity and warm, but not sugary or shampoo-y or headache-inducing. - This is more of a sensation than a scent. - The lavender isn't sharp or too aromatic. - It is almost a "balm" effect more than a perfume. - But not in a vanilla or amber sort of way.
Theme W4: Within-Category Recombination (Creamy Gourmand Accords)
- Slight hint of vanilla rounds it out. - It's like a caramel vanilla. - The vanilla adds the sweet note. - It has honey like underlying something as well. - There is also a subtle sweetness in the background (vanilla?), - A bit of creaminess is added from the vanilla. - A warm, sweet, slightly powdery and sensual. - Yet it has a lovely creaminess to it. - A bit sweet, a bit creamy. - It also has sort of almond qualities to it.
Theme W5: Within-Category Recombination (Coconut Gourmand Accords)
- There is a little hint of coconut. - There is also coconut. - This is a very vanillary coconut scent. - To me there is a faint scent of coconut. - Coconut is one of my favorites. - There is a savory coconut effect lurking just around the edges. - I sense a little coconut in there but not much. - The coconut adds a creaminess. - This is a pretty decent coconut scent. - In this perfume, I also get a smell of coconut.

Table K.6: Themes Distinguishing Atypical from Typical Women’s Fragrances (Confirmatory Sample)

Theme W6: Cross-Gender Recombination (Spicy and Resinous Accords)
- Almost gourmand by scent. - Incense-y, herbal, but also floral. - There’s an herbal feel to it. - There is a hint of spice in the background. - There’s also an incense vibe to it, cinnamon, resins and balsams are strong along with amber, too. - To my nose this is citrusy vanilla with some woody touch. - This is a gourmand scent for them. - Sweet spices radiating from within. - This is an aromatherapy piece. - This scent is almost magical in its complexity and balance.
Theme W7: Novelty and Distinctiveness
- There is something intoxicating about this scent. - This an interesting scent. - It’s an intoxicating scent. - The scent is intoxicating. - This smells like candy. - Smells cakey or like a dessert. - The scent is very juicy. - It is as if something a tad aromatic accompanies it. - It appears something that I call: marjoram scent. - It is an aura of a scent.
Theme W8: Performance Intensity and Wearability
- It last a long time on the skin. - It lasts a long time. - It’s strong and long lasting. - Lasts quite a long time on the skin. - It has a long life span on the skin. - Not overwhelming but it lasts a very long time on the skin. - It stays on your skin for a long time. - It does last quite well on the skin. - It’s potent and lasts for hours on the skin. - Lasts many hours on skin, and has a noticeably pretty sillage.
Theme W9: Gender Boundary Blurring (Unisex Perception)
- It’s a rather unisex scent. - It is def a unisex fragrance. - This is also a unisex fragrance. - This does have a unisex vibe. - This one can be a unisex scent. - This could be a unisex scent. - It can pass for a unisex scent. - Definitely a unisex fragrance. - It is Unisex in my opinion. - Definitely a unisex scent.
Theme W10: Within-Category Recombination (Floral Intensification)
- I really like the tuberose in this. - A gentle tuberose scent. - If you like tuberose you will like this perfume. - Might be the perfect tuberose fragrance imo. - The Tuberose isn’t as harsh as it is in some other fragrances. - I think I detect a hint of tuberose in the background. - It’s the tuberose that stands out most to my nose. - Interesting perfume with a clearly floral soul, centered on tuberose. - I love Tuberose fragrances. - Love the tuberose in the formula.

Table K.7: Themes Distinguishing Atypical from Typical Women's Fragrances (Confirmatory Sample)

Theme W11: Perceived Artistry and Refined Appeal
- There's something magical about it. - A high quality composition that has its own unique charms. - It is absolutely lovely. - It is truly lovely and quite unusual. - My God, it's gorgeous. - It's one of a kind. - This is a stunning product. - This is simply stunning. - There is something magical, deep and woody in this. - It is actually quite refined.
Theme W12: Intense Positive Affect and Enthusiastic Evaluation
- I adore this to bits. - I just adore this. - I find this lovely. - To me, this is just lovely. - I simply adore this. - I feel lucky to have accidentally discovered this Guerlain gem. - One of my new favourites. - This is almost a "love" for me but since I am in my gourmand phase I'll just say I like it a lot! - I LIKE this so much! - I think this is lovely.
Theme W13: Within-Category Recombination (White Floral Intensification)
- Gardenia and jasmine stands out. - This is a gourmand floral. - One of the best oriental florals. - It's floral with a slightly gourmand background. - A beautiful airy white floral. - It has a soft gourmand-floral feel. - An engaging white floral with a little mineral twist. - This is a white floral that I can adore. - A fresh green floral. - One of the most gorgeous white florals I've experienced.
Theme W14: Evocative Sensory Appeal and Hedonic Richness
- This scent is delicious. - This is a delectable scent. - It is a delicious fragrance. - What a delicious fragrance. - This is such a delicious scent. - This is a beautiful gourmand scent. - This is a fantastic fragrance. - This scent is amazing. - This scent smells delicious. - This is a true masterpiece of a scent.
Theme W15: Accord-Profile Disconfirmation
- That is the drawback of this scent for me. - there's way too much amber in this, it smells very savoury and warm and yellow.. I prefer vanilla to be sweet or boozy. - I think it is the amber and vanilla making it not my cup of tea. - I don't get the vanilla that some people smell in this. - I also don't understand the macaron note, macarons are made of almonds and I get nothing like that from this scent. - I get also a patchouli vibe, but I don't believe there is any patchouli in it, otherwise I would hate it. - I don't see this as a fruity floral but rather a spicy (oriental) floral. - This pulls surprisingly gourmand on my skin rather than floral. - it doesn't smell like white chocolate for me :(- I find this to be waayyy to pungent, the French pastries note is overpoweringly sickly, it's the only thing I can smell 3/10

Table K.8: Themes Distinguishing Atypical from Typical Women’s Fragrances (Confirmatory Sample)

Theme W16: Ownership, Sampling, and Comparative Experiences
- Edit: does anybody else see the resemblance to L’Heure Bleue of Guerlain (without the anise and carnation spiciness?) - If anyone has a recommendation for something that smells just like this, but better longevity and projection - please let me know! - I think that is a good substitute for Infusion d’Iris Absolue Prada. :) - I hope I will have a chance to try also eau de parfum version - How I wish someone would create an eau de Parfum of this! - I Wold Like To Test This one I See That Alot Of People Love This perfume So I Think I will To :) XD XD XD - Reminds me a lot of the l’eclat flanker but I really enjoy all of the LVEB releases for an easy throw on perfume. - Just wanted to add that Caress makes a body wash called Sheer Twilight that smells almost just like Seduction Dark Orchid. - Edit: If you enjoy the scent on your skin, go ahead and buy it! - An eau de parfum version would be nice perhaps?
Theme W17: Ambivalence and Qualified Evaluation
- I have mixed feelings on this one. - I have mixed feelings on this little gem. - This is one of the latter. - I HAD A SMALL SAMPLE OF THIS. - I’m mixed on this one. - Just want to come here after trying again and again to like this to just say that this isn’t a safe blind buy. - I am a bit torn with my feelings of this one. - I had a sample of this. - Please try before you buy. - This one is the latter.

Table K.9: Themes Distinguishing Typical from Atypical Unisex Fragrances (Confirmatory Sample)

Theme U1: Wearability and Experiential Appeal
- This could be worn in autumn, winter or spring. - It's warm but has some different facets. - It feels good to have on. - I have received several compliments. - I have gotten several compliments. - This is made for autumn but would do well in winter and spring as well. - The more I wear it the more I like it. - It is crisp yet pleasant. - Anyone can wear this. - Good for many occasions.
Theme U2: Balanced Olfactory Recombination (Warm and Resinous Accords)
- There are woody and warm, slightly powdered (amber) sweet touches, too. - There's a touch of powdery sandalwood. - The base is a wood and there is some spice to it as well. - A amber>vanilla>lemon scent with some spices. - Very well blended woody, slightly sweet vanilla kind of scent. - It is gently spiced with the addition of sandalwood. - I love the fruity/incense/sandalwood combo. - This is like an incensey sandalwood. - The base notes definitely fit the cozy vibe of the whole fragrance, soft vanilla and sandalwood being the main thing I detect. - Light tobacco and sandalwood compliment this.
Theme U3: Ownership, Sampling, and Value-Based Evaluation
- Obviously the price is quite high so sample before you buy - Price-tag is high, but not ridiculous. (- A bit on the pricey side so sample first. - On a side note & I mean no harm but it's worth clarifying that this fragrance is only \$140 US dollars per 100mL not \$200. - I'm glad I tried it but you need something much more special for this price. - Could the price tag and brand presense be a factor? - P.S. If you find this way too overpriced, I highly recommend Fragrance World's 'Joyness Oud'. - For a perfume that's basically extrait I would've liked it if it lasted longer but it's €48 for 100ml, I'm not gonna complain. - But the price tag is not equally likeable. - I will keep and use my decant but I'm not convinced to pay the full price for this (despite the fact that it is well blended etc.).
Theme U4: Balanced Olfactory Recombination (Woody and Aromatic Accords)
- Very incense spicy woody fragrance. - There is an incense vibe. - Woody, fruity, resinous scent with an incense undertone. - A deeply woody fragrance with a pop of fresh spice. - This fragrance has a lot of greens and woods in it. - Spicy, woody and a bit resinous. - The smoke has a hint of incense about it. - The woody scent with a small amount of vanilla gets my attention. - It smells familiar but also unique at the same time. - Spicy aromatic woody floral.
Theme U5: Perceived Artistry and Distinctive Appeal
- This fragrance is a beauty. - This is a beautiful fragrance. - A very nice fragrance. - The fragrance is beautiful. - This is an interesting fragrance. - The fragrance is gorgeous. - This is a gorgeous fragrance. - This fragrance is beautiful. - Its a beautiful fragrance. - It's a beautiful fragrance.

Table K.10: Themes Distinguishing Typical from Atypical Unisex Fragrances (Confirmatory Sample)

Theme U6: Qualified Evaluation and Mild Disappointment
- I like this, but I don't love it. - It is just not in the same league. - I think this one falls in the "I appreciate the artistry, but don't actually like it that much" category. - My first impression is not great due to performance alone. - Performance is not the best. - I think this is a bit overhyped. - It's not groundbreaking but it's just a little bit different. - Not sure I have any good direct comparisons for this one. - To me, this is not intoxicating. - Performance is not as good as people claim.
Theme U7: Perceived Artistry and Compositional Excellence
- This is a beauty. - It's very well blended and unique. - This is very well done. - A thing of beauty. - The performance is outstanding. - The performance is remarkable. - This is excellent and refined. - It is masterfully blended. - Other people have noted that the sillage is fantastic in this one. - It's clean and heavy at the same time.
Theme U8: Gender Perception (Directional Lean)
- Unisex, leans masculine in my opinion. - It is unisex, maybe leans a bit feminine. - This leans masculine, but is unisex. - It feels unisex but it leans slightly to masculine scents. - This is more masculine than unisex. - It is unisex, but leans a tad bit feminine. - More masculine than unisex, in my opinion. - This is a unisex scent, leaning perhaps a smidge towards masculine. - Unisex but leans masculine. - It is unisex, but leans a bit towards female I think.
Theme U9: Balanced Olfactory Recombination (Oud Integration)
- The oud is nice. - Very nice twist on an Oud fragrance. - The oud isn't too loud and well blended into this fragrance. - This is an oud rose fragrance after all. - If you are an oud fan this one is a good one to try. - It is an Oud fragrance/ A good Oud fragrance! - The Oud in this is on the woody side and a nice one. - This is how an Oud perfume should be. - The Oud is there, yet it's not overwhelming. - Socially acceptable Oud fragrance.
Theme U10: Negative Expectation Disconfirmation (Conceptual Ambiguity)
- A good but not astonishing fragrance. - It's not the magnanimous fragrance I expected. - I like this fragrance, but I'm not the biggest fan of sweet scents. - This one is nice, but not an original scent. - It's indeed not the most refined fragrance. - Not a formal fragrance...but not a completely casual fragrance either. - This is not a harmonious fragrance. - However, this is not an "oudy" fragrance. - Still, this fragrance isn't bad IMPO...its just not original nor as good as those two aforementioned fragrances since I smelled those first. - I have to say, this isn't the best perfume of its category.

Table K.11: Themes Distinguishing Typical from Atypical Unisex Fragrances (Confirmatory Sample)

Theme U11: Balanced Olfactory Recombination (Temporal Evolution) - In the drydown the whole scent gets woody and somehow sweeter. - As it moves further into dry-down a slight sweetish floral tone comes out in the woodiness. - The dry-down has an incense vibe. - Dries down to a fruity, earthy scent. - As the fragrance dries down there are some woods coming out to balance the sweetness out. - Dries down to a subtle woody smell. - The drydown has some slightly dry woods and musk to add to the mix. - During the dry down the wood, spice and slightly floral notes become more pronounced. - In the final drydown incenseish but not quite resinous enough. - I imagine there's elemi, sandalwood and other woody, resinous notes emerging, which contribute to the pleasant dry down and base.
Theme U12: Balanced Olfactory Recombination (Layered Integration) - Opens with a rather nice citrus note. - Opens with a slight fresh citrus along with some herbal greens thrown in along with a touch of mint. - Starts off very citrus but also, almost gingery. - Opens citrusy, green and slightly herbal. - There's a bit of spice emerging as well. - This is a sophisticated citrus. - Citrusy and green in the opening. - The opening is a very zesty, tangy citrus with a hint of pepper. - There are additional spices and aromatic notes, especially in the top. - The opening is citrus and spices, mingling with a pretty frankincense.
Theme U13: Performance Intensity and Wearability - 8+ hours longevity, moderate projection. - Longevity is around 8 hours. - Longevity is around 8 hours on me and the sillage is good. :) - Performance is decent, approx 6-8 hours with mild projection. - Longevity is good, I get around 5 hours with moderate sillage. - 12 hours longevity with a light, but noticeable Sillage. - The performance is very good, about 8-9 hours longevity some times a bit more and good projection for 2-3 hours. - Can go up to 8 hours longevity, explosive on the first 2-3 hours but fades a bit. - 6-8 hour longevity and noticeable sillage.) - Soft projection for 2-3hr, longevity 6-8hr.
Theme U14: Comparative Evaluation (Scent Similarity) - Not exactly Aventus-esque but the scent profile definitely reminds me of it. - Not exact dupes but similar vibes. - Something about it kinda reminds me of D&G Pour Homme. - But after that it is very similar to the mentioned perfume. - Not a dupe, but it shares the same vibe to an extent. - Reminds me alot and similar too Swiss Arabian Shaghaf oud. - It's almost like a gourmand bergamot scent. - IMO - this has some resemblance to BDK Gris Charnel Extrait (and the EDP one). - There is indeed a slight resemblance to Cologne pour le Soir, this one leans more gourmand though. - Reminds me of Xerjoff renaissance except the petitgrain is a little harsher and theres a slight fruity floral note underneath everything.
Theme U15: Ownership, Sampling, and Comparative Experiences - P/s: If you're looking for a tobacco scent with beastly performance, then try Al Haramain's Amber Tobacco Edition or TF Tobacco Vanille (you wont regret) - PS: I sampled Orion by Terrenzi, and it smells exactly like Rasana @perfumista_225 for Middle eastern perfume reviews. - UPDATE: The vanilla notes in this one remind me of the same vanilla notes I get from Molinard's Vanille Fruitee...if this helps anyone consider on this fragrance! - Edit: I ordered a FB of this and it is slowly becoming one of my favorite fragrances. - longevity is decent too.cold weather fragrance btw 8/10 - But for my next neroli fragrance im considering bvlgari wood neroli - Sillage: 7./10 Longevity: 8.5/10 Scent: 9./10 Overall: 8.5/10 - Performance seems good from the sample I tried; good longevity (especially for a CdG) and projection seems appropriate for such a heavy fragrance (ie. Just to show i'm not trying to hate on Creed, i've got Aventus and Royal Oud decants as well and they are utterly beautiful scents. - Scent: 8/10 Longevity: 10/10 Sillage: 8/10 Cost ratio: 4/10 Overall: 7.5/10 I'm torn because I could have liked this fragrance yet something is missing.

L Web Appendix L: Theme Correlations and Complete Theme-Level Results

To assess the extent to which the extracted themes capture distinct semantic dimensions, we examine pairwise correlations among theme indicators separately for men’s, women’s, and unisex fragrances. Tables L.1–L.3 report the corresponding correlation matrices. Correlations are uniformly small across all three segments, with the largest pairwise correlation equal to 0.097. These results indicate limited overlap among themes, suggesting that they capture distinct aspects of review content with little redundancy. Finally, Table L.4 presents the full results for all themes from both the descriptive and controlled analyses.

Table L.1: Correlation Matrix of Themes for Men Fragrances

	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13
M1	1.000												
M2	0.013	1.000											
M3	0.026	0.044	1.000										
M4	0.011	0.043	0.052	1.000									
M5	0.009	0.011	0.044	0.021	1.000								
M6	0.006	0.030	0.071	0.029	0.019	1.000							
M7	0.032	0.013	0.058	0.026	0.040	0.037	1.000						
M8	0.052	0.026	0.070	0.051	0.047	0.062	0.044	1.000					
M9	0.023	0.036	0.054	0.051	0.034	0.042	0.056	0.068	1.000				
M10	0.036	0.047	0.033	0.047	0.045	0.026	0.050	0.059	0.097	1.000			
M11	0.009	0.001	0.050	0.016	0.034	0.017	0.050	0.023	0.026	0.035	1.000		
M12	0.035	0.046	0.054	0.043	0.016	0.049	0.074	0.051	0.076	0.069	0.029	1.000	
M13	-0.000	0.011	0.038	0.037	0.055	0.027	0.043	0.048	0.045	0.035	0.006	0.019	1.000

Table L.2: Correlation Matrix of Themes for Women Fragrances

	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16	W17
W1	1.000																
W2	0.012	1.000															
W3	0.037	0.011	1.000														
W4	0.035	-0.003	0.035	1.000													
W5	0.010	-0.005	0.002	0.042	1.000												
W6	0.081	0.029	0.045	0.018	-0.008	1.000											
W7	0.055	0.007	0.040	0.047	0.020	0.036	1.000										
W8	0.020	0.001	0.021	0.018	0.010	0.016	0.023	1.000									
W9	0.048	0.023	0.019	0.005	-0.004	0.055	0.017	0.012	1.000								
W10	0.016	-0.004	0.027	-0.002	0.000	0.001	0.014	0.015	0.001	1.000							
W11	0.042	0.029	0.026	0.014	-0.000	0.063	0.019	0.011	0.022	0.034	1.000						
W12	0.025	0.014	0.016	0.035	0.012	0.036	0.027	0.015	0.004	0.006	0.025	1.000					
W13	0.027	0.016	0.031	0.003	-0.001	0.027	0.024	0.015	0.004	0.076	0.026	0.011	1.000				
W14	0.053	0.020	0.032	0.036	0.016	0.053	0.038	0.028	0.024	0.005	0.047	0.027	0.054	1.000			
W15	0.024	0.002	0.051	0.022	0.034	0.024	0.031	0.007	0.005	0.014	0.013	0.005	0.022	0.005	1.000		
W16	0.026	0.013	0.017	0.019	0.002	0.018	0.017	0.013	0.016	0.001	0.017	0.022	0.004	0.027	0.008	1.000	
W17	0.024	0.006	0.022	0.010	-0.008	0.029	0.010	0.012	0.020	-0.000	0.013	0.024	0.004	0.011	0.007	0.019	1.000

Table L.3: Correlation Matrix of Themes for Unisex Fragrances

	U1	U2	U3	U4	U5	U6	U7	U8	U9	U10	U11	U12	U13	U14	U15
U1	1.000														
U2	0.020	1.000													
U3	0.009	0.014	1.000												
U4	0.011	0.063	0.004	1.000											
U5	0.044	0.070	0.031	0.045	1.000										
U6	0.022	0.025	0.036	0.017	0.021	1.000									
U7	0.036	0.051	0.036	0.037	0.085	0.039	1.000								
U8	0.023	0.027	0.022	0.043	0.041	0.017	0.040	1.000							
U9	0.018	0.026	0.020	0.010	0.020	0.039	0.036	0.016	1.000						
U10	0.019	0.022	0.024	0.028	0.041	0.054	0.026	0.015	0.036	1.000					
U11	0.016	0.047	0.013	0.059	0.027	0.022	0.022	0.027	0.007	0.000	1.000				
U12	0.023	0.078	0.014	0.055	0.057	0.029	0.034	0.042	0.012	0.021	0.041	1.000			
U13	0.019	0.012	0.023	0.013	0.038	0.012	0.029	0.019	0.022	0.023	0.024	0.017	1.000		
U14	0.003	0.016	0.029	0.017	0.025	0.033	0.040	0.012	0.018	0.015	0.012	0.014	0.026	1.000	
U15	0.020	0.016	0.022	0.025	0.017	0.033	0.005	0.011	0.028	0.014	0.012	-0.002	0.022	0.026	1.000

Table L.4: Theme Prevalence and Controlled Associations with Atypicality

Theme	Descriptive Analysis				Controlled Analysis		
	Q_4	Q_1	Δ	p_{diff}	β	SE	p_{reg}
Men							
M1: Seasonal Suitability and Cold-Weather Appeal	0.019	0.010	0.009	< 0.001	0.052	0.017	0.003
M2: Material-Centric Recombination (Oud)	0.040	0.007	0.033	< 0.001	0.132	0.037	< 0.001
M3: Intense Positive Evaluation and Aesthetic Appeal	0.107	0.061	0.046	< 0.001	0.032	0.010	0.001
M4: Material-Centric Recombination (Leather)	0.074	0.012	0.062	< 0.001	0.107	0.015	< 0.001
M5: Ownership, Sampling, and Acquisition Experiences	0.053	0.036	0.018	< 0.001	0.008	0.008	0.323
M6: Hedonic Richness and Emotional Appeal	0.051	0.028	0.024	< 0.001	0.035	0.010	< 0.001
M7: Cross-Gender Recombination (Sweet and Gourmand Accords)	0.070	0.029	0.041	< 0.001	0.035	0.009	< 0.001
M8: Intensity and Sensory Presence	0.065	0.037	0.028	< 0.001	0.038	0.008	< 0.001
M9: Idiosyncratic Olfactory Impressions and Sensory Ambiguity	0.074	0.038	0.036	< 0.001	0.036	0.010	< 0.001
M10: Negative Expectation Disconfirmation and Taste-Based Rejection	0.061	0.030	0.031	< 0.001	0.042	0.010	< 0.001
M11: Gender Boundary Blurring (Unisex Perception)	0.027	0.018	0.009	< 0.001	0.003	0.016	0.863
M12: Within-Category Recombination (Tobacco and Smoky Accords)	0.066	0.018	0.048	< 0.001	0.063	0.014	< 0.001
M13: Ambivalence and Taste-Based Uncertainty	0.048	0.033	0.015	< 0.001	0.023	0.011	0.036
Women							
W1: Evocative Depth and Atmospheric Complexity	0.041	0.018	0.022	< 0.001	0.073	0.009	< 0.001
W2: Within-Category Recombination (Rose Intensification)	0.024	0.008	0.016	< 0.001	0.012	0.018	0.501
W3: Negative Expectation Disconfirmation (Category Ambiguity)	0.029	0.014	0.015	< 0.001	0.051	0.014	< 0.001
W4: Within-Category Recombination (Creamy Gourmand Accords)	0.055	0.022	0.033	< 0.001	0.058	0.012	< 0.001
W5: Within-Category Recombination (Coconut Gourmand Accords)	0.026	0.005	0.020	< 0.001	0.114	0.026	< 0.001
W6: Cross-Gender Recombination (Spicy and Resinous Accords)	0.045	0.020	0.026	< 0.001	0.089	0.014	< 0.001
W7: Novelty and Distinctiveness	0.073	0.034	0.039	< 0.001	0.053	0.008	< 0.001
W8: Performance Intensity and Wearability	0.021	0.017	0.005	< 0.001	0.007	0.011	0.517
W9: Gender Boundary Blurring (Unisex Perception)	0.012	0.005	0.007	< 0.001	0.083	0.020	< 0.001
W10: Within-Category Recombination (Floral Intensification)	0.033	0.007	0.026	< 0.001	0.029	0.022	0.197
W11: Perceived Artistry and Refined Appeal	0.036	0.019	0.016	< 0.001	0.054	0.009	< 0.001
W12: Intense Positive Affect and Enthusiastic Evaluation	0.041	0.023	0.017	< 0.001	0.039	0.009	< 0.001
W13: Within-Category Recombination (White Floral Intensification)	0.035	0.015	0.020	< 0.001	-0.021	0.008	0.009
W14: Evocative Sensory Appeal and Hedonic Richness	0.071	0.038	0.033	< 0.001	0.022	0.007	< 0.001
W15: Accord-Profile Disconfirmation	0.035	0.017	0.018	< 0.001	0.059	0.010	< 0.001
W16: Ownership, Sampling, and Comparative Experiences	0.034	0.018	0.016	< 0.001	0.041	0.010	< 0.001
W17: Ambivalence and Qualified Evaluation	0.026	0.017	0.010	< 0.001	0.034	0.011	0.002
Unisex							
U1: Wearability and Experiential Appeal	0.024	0.034	-0.010	< 0.001	0.001	0.011	0.966
U2: Balanced Olfactory Recombination (Warm and Resinous Accords)	0.028	0.072	-0.043	< 0.001	-0.062	0.011	< 0.001
U3: Ownership, Sampling, and Value-Based Evaluation	0.016	0.029	-0.013	< 0.001	-0.026	0.012	0.032
U4: Balanced Olfactory Recombination (Woody and Aromatic Accords)	0.042	0.076	-0.034	< 0.001	-0.043	0.012	< 0.001
U5: Perceived Artistry and Distinctive Appeal	0.081	0.127	-0.046	< 0.001	-0.015	0.006	0.012
U6: Qualified Evaluation and Mild Disappointment	0.034	0.044	-0.010	< 0.001	0.013	0.010	0.220
U7: Perceived Artistry and Compositional Excellence	0.038	0.063	-0.024	< 0.001	0.005	0.009	0.546
U8: Gender Perception (Directional Lean)	0.034	0.054	-0.020	< 0.001	-0.015	0.009	0.086
U9: Balanced Olfactory Recombination (Oud Integration)	0.018	0.025	-0.007	0.003	-0.011	0.015	0.486
U10: Negative Expectation Disconfirmation (Conceptual Ambiguity)	0.041	0.061	-0.020	< 0.001	-0.023	0.011	0.031
U11: Balanced Olfactory Recombination (Temporal Evolution)	0.021	0.035	-0.014	< 0.001	-0.032	0.013	0.010
U12: Balanced Olfactory Recombination (Layered Integration)	0.025	0.049	-0.024	< 0.001	-0.032	0.010	0.002
U13: Performance Intensity and Wearability	0.016	0.020	-0.003	0.079	0.007	0.014	0.620
U14: Comparative Evaluation (Scent Similarity)	0.025	0.047	-0.022	< 0.001	-0.013	0.010	0.219
U15: Ownership, Sampling, and Comparative Experiences	0.026	0.040	-0.015	< 0.001	-0.013	0.009	0.162

Note. Q_1 and Q_4 denote the lowest and highest atypicality quartiles, respectively. Columns Q_4 and Q_1 report average review-level theme prevalence for fragrances in the corresponding quartiles. Δ reports the difference in mean theme prevalence ($Q_4 - Q_1$), and p_{diff} reports the corresponding two-sample difference-in-means test. Columns β , SE, and p_{reg} report coefficients, standard errors, and p -values from segment-specific linear probability models in which membership in Q_4 is regressed on review-level theme prevalence measures and the full set of controls described in the text. For the unisex segment, negative coefficients indicate themes associated with the more typical fragrances (Q_1). Standard errors are clustered at the brand level for the linear probability model.

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